Towards Braided ∞-Operads

by

Franziska Ahrens

born

03.07.1997

A thesis submitted to the Department of Mathematics

Universität Hamburg

In Partial Fulfillment of the Requirements

for the Degree of Master of Science

June 18, 2025

Supervisor: Prof. Dr. Tobias Dyckerhoff

Examiner: Prof. Dr. Julian Holstein

Contents

1 Operads

Operads are a concept that originally occurred when studying loop spaces. Let ΩX be the space of based loops in a topological space X. Its structure reflects an interesting form of algebraic composition. The composition of loops is only associative up to homotopy equivalence meaning that loop spaces behave like a monoid in a homotopy-theoretic sense, but not in a strict algebraic sense. In algebraic topology there is a need for a framework that formalizes composition up to homotopy, which is where operads come in. An operad can be thought of as a structure that encodes a family of operations, each of which has a certain number of inputs, and a way to compose these operations together. The composition of operations in an operad satisfies associativity and unit laws.
Let $\alpha\colon [0,1] \to X$ and $\beta\colon [0,1] \to X$ be loops in X having the same base point x_0. Their composition $\beta \circ \alpha$ is usually defined by splitting the unit interval in half, i.e.

$$\beta \circ \alpha(s) := \begin{cases} \alpha(2s) & \text{for } s \in [0, \frac{1}{2}] \\ \beta(2s-1) & \text{for } s \in (\frac{1}{2}, 1]. \end{cases}$$

This concatenation in itself is not associative; consider $\alpha, \beta, \gamma \in \Omega X$. Then by the given definition $\gamma \circ (\beta \circ \alpha)$ and $(\gamma \circ \beta) \circ \alpha$ are different maps:

$$\gamma \circ (\beta \circ \alpha)(s) = \begin{cases} \alpha(4s) & \text{for } s \in [0, \frac{1}{4}] \\ \beta(4s-1) & \text{for } s \in (\frac{1}{4}, \frac{1}{2}] \\ \gamma(2s-1) & \text{for } s \in (\frac{1}{2}, 1] \end{cases}$$

$$(\gamma \circ \beta) \circ \alpha(s) = \begin{cases} \alpha(2s) & \text{for } s \in [0, \frac{1}{2}] \\ \beta(4s-2) & \text{for } s \in (\frac{1}{2}, \frac{3}{4}] \\ \gamma(4s-3) & \text{for } s \in (\frac{3}{4}, 1] \end{cases}$$

The choice of splitting the unit intervall in half is merely a convention. Hence, composition of loops can be thought of as splitting intervals in an arbitrary manner; or in other words embedding unit intervals into a unit interval such that the image of the inner intervals are disjoint.

$$\big[[I_1] \dots [\quad I_i \quad] \dots [\, I_n \,]\big]$$

So, for an ordered family of loops $\{\alpha_i\colon I_i \to X\}_{1 \leq i \leq n}$ there is for each order preserving concatenation β a unique map $f \in \mathrm{Emb}(I_1, \dots, I_n, I)$ such that $\beta \circ f(I_i) = \alpha_i$ for all $1 \leq i \leq n$. We denote the set of embeddings by $\mathrm{E}(n) = \mathrm{Emb}(I_1, \dots, I_n, I)$ for each $n \in \mathbb{N}$. For every $(f_k, (f_{j_i})_{1 \leq i \leq k}) \in \mathrm{E}(k) \times \mathrm{E}(j_1) \times \dots \times \mathrm{E}(j_k)$ we obtain an embedding $f \in \mathrm{E}(j_1 + \dots + j_k)$ by defining $f = f_k \circ (f_{j_1} \sqcup \dots \sqcup f_{j_k})$. This defines a composition map

$$\gamma\colon \mathrm{E}(k) \times \mathrm{E}(j_1) \times \dots \times \mathrm{E}(j_k) \to \mathrm{E}(j_1 + \dots + j_k)$$

Since the composition of maps is associative the following diagram commutes:

$$\begin{tikzcd}
\mathrm{E}(k) \otimes (\prod_{s=1}^k \mathrm{E}(j_s)) \times (\prod_{r=1}^j \mathrm{E}(i_r)) \arrow[r, "\gamma \otimes \mathrm{Id}"] \arrow[dd, "s"] & \mathrm{E}(j) \times (\prod_{r=1}^j \mathrm{E}(i_r)) \arrow[d, "\gamma"] \\
& \mathrm{E}(i) \\
\mathrm{E}(k) \otimes (\prod_{s=1}^k (\mathrm{E}(j_s) \times \prod_{q=1}^{j_s} \mathrm{E}(i_{g_{s-1}+q})) \arrow[r, "\mathrm{Id} \times \gamma^s"'] & \mathrm{E}(k) \times ((\prod_{s=1}^k \mathrm{E}(h_s)) \arrow[u, "\gamma"']
\end{tikzcd}$$

where s shuffles the cartesian product and

$$n,\ j_1, \dots, j_k,\ i_1, \dots, i_j \in \mathbb{N}$$

with
$$\sum_{1\leq s\leq k} j_s = j,\ \sum_{1\leq r\leq j} i_r = i,\ g_s = \sum_{0\leq l\leq s} j_l \text{ and } h_s = \sum_{1\leq q\leq j_s} i_{g_{s-1}+q}.$$
Further composing with the identity embedding in E(1) does nothing. Thus, we have seen our first example of an operad, providing a framework in which the composition of loop spaces satisfies associativity.
In this way, operads provide a more general and flexible formalism for capturing the algebraic structures found in loop spaces and other topological constructs where composition occurs up to homotopy. Similarly to monoids that can be considered as single object categories, operads can be considered as single object multicategories. In this section, we will introdue the notion of a planar and a symmetric operad as they were introduced by May in [KM95]. Then, we will move on to the more general notion of a colored operad following [LJ]. We will establish a second equivalent definition of a colored operad that is purely functorial, meaning we can therefore also phrase it in an ∞-categorial context. In the next section we will then consider braided operads and we work towards defining braided colored operads. Ultimately, our goal is a purely functorial definition of a braided colored operad such that we can lift the concept to an ∞-categorial context as well.

1.1 Planar and Symmetric Operads

From now on, unless it is stated otherwise, let $\mathcal{C}$ be a symmetric monoidal category with tensor product $\otimes$ and tensor unit I. For an object C in $\mathcal{C}$, we write C^j for the j-th tensor power of C.

Definition 1.1. A *planar operad* $\mathcal{P}$ in $\mathcal{C}$ consists of a family of objects $(\mathcal{P}(j))_{j\in\mathbb{N}}$ together with a unit morphism $\eta\colon k\to\mathcal{P}(1)$ and a product morphism
$$\gamma\colon \mathcal{P}(k)\otimes\mathcal{P}(j_1)\otimes\cdots\otimes\mathcal{P}(j_k)\to\mathcal{P}(j),$$
where $\sum_{1\leq i\leq k} j_i = j$ such that the following associativity and unitality constraints hold:

- For any choice of integers
$$n,\ j_1,\dots,j_k,\ i_1,\dots,i_j$$
with
$$\sum_{1\leq s\leq k} j_s = j,\ \sum_{1\leq r\leq j} i_r = i,\ g_s = \sum_{0\leq l\leq s} j_l \text{ and } h_s = \sum_{1\leq q\leq j_s} i_{g_{s-1}+q}$$
the following diagram commutes:
$$\begin{tikzcd}
\mathcal{P}(k)\otimes(\bigotimes_{s=1}^k\mathcal{P}(j_s))\otimes(\bigotimes_{r=1}^j\mathcal{P}(i_r)) \arrow[r, "\gamma\otimes\mathrm{Id}"] \arrow[dd, "s"] & \mathcal{P}(j)\otimes(\bigotimes_{r=1}^j\mathcal{P}(i_r)) \arrow[d, "\gamma"] \\
& \mathcal{P}(i) \\
\mathcal{P}(k)\otimes(\bigotimes_{s=1}^k(\mathcal{P}(j_s)\otimes\bigotimes_{q=1}^{j_s}\mathcal{P}(i_{g_{s-1}+q})) \arrow[r, "\mathrm{Id}\otimes(\otimes_s\gamma)"'] & \mathcal{P}(k)\otimes((\bigotimes_{s=1}^k\mathcal{P}(h_s)) \arrow[u, "\gamma"']
\end{tikzcd}$$
where s shuffles the tensor product.

- For all $j\in\mathbb{N}$ the following diagrams commute:
$$\begin{tikzcd}
\mathcal{P}(j)\otimes k^j \arrow[r, "\cong"] \arrow[d, "\mathrm{Id}\otimes(\otimes_j\eta)"'] & \mathcal{P}(j) \\
\mathcal{P}(j)\otimes\mathcal{P}(1)^j \arrow[ur, "\gamma"'] &
\end{tikzcd}
\qquad
\begin{tikzcd}
k\otimes\mathcal{P}(j) \arrow[r, "\cong"] \arrow[d, "\eta\otimes\mathrm{Id}"'] & \mathcal{P}(j) \\
\mathcal{P}(1)\otimes\mathcal{P}(j) \arrow[ur, "\gamma"'] &
\end{tikzcd}$$

Examples 1.2. 1. Later we will define what it means to be an algebra over an operad. The *associative operad* $\mathrm{Ass}^{\otimes}$ is the operad whose algebras are monoids. The associative planar operad in $\mathcal{C}$ is the operad ${\mathrm{Ass}^{\otimes}}_{\mathcal{C}}(n) = \mathrm{I}$ for all $i \in \mathbb{N}$ with Id_{I} being the unit and the composition map being the canonical isomorphism. Hence, for $\mathcal{C} = \mathbf{Set}$, the category of small sets equipped with the cartesian product, we obtain exactly one n-ary operation for each n: ${\mathrm{Ass}^{\otimes}}_{\mathbf{Set}}$ is defined by ${\mathrm{Ass}^{\otimes}}_{\mathbf{Set}}(n) = *$ for all n. Similarly, for $\mathcal{C} = \mathbf{Vect}_K$ we get the free abelian vector space over $*$, i.e. K.

2. At the beginning of the section we already saw that the concatenation of loops defines an operad: Let $\mathcal{C} = \mathbf{Top}$. Then we can define the *operad of little interval* E as

$$\mathrm{E}(n) = \mathrm{Emb}(I_1, \dots, I_n, I),$$

the set of orientation-preserving embeddings of n disjoint intervals into the unit interval I. We equip $\mathrm{E}(n)$ with the subspace topology from being a subset of $\mathbb{R}^n$. Composition is defined by inserting the intervals into another interval as we have seen above.

3. For $\mathcal{C}$ being closed, i.e. having an internal hom functor $\underline{\mathrm{Hom}}(-,-)$, we can define another planar operad End_X for any object $X \in \mathcal{C}$:

$$\mathrm{End}_X(n) \coloneqq \underline{\mathrm{Hom}}(X^{\otimes n}, X).$$

The unit morphism η is given by the image of Id_X under the adjunction:

$$\mathrm{Hom}(X, X) \cong \mathrm{Hom}(I, \underline{\mathrm{Hom}}(X, X)).$$

Further, the composition morphism

$$\underline{\mathrm{Hom}}(X^{\otimes k}, X) \otimes \bigotimes_s \underline{\mathrm{Hom}}(X^{\otimes j_s}, X) \to \underline{\mathrm{Hom}}(X^{\otimes j}, X)$$

is given by composition:

$$\begin{array}{c} \underline{\mathrm{Hom}}(X^{\otimes k}, X) \otimes \bigotimes_s \underline{\mathrm{Hom}}(X^{\otimes j_s}, X) \\ \downarrow \\ \underline{\mathrm{Hom}}(X^{\otimes k}, X) \otimes \underline{\mathrm{Hom}}(X^{\otimes j}, X^{\otimes k}) \\ \downarrow \\ \underline{\mathrm{Hom}}(X^{\otimes j}, X) \end{array}$$

We call End_X the *endomorphism operad over* X.

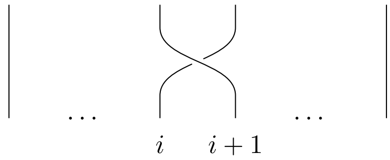
For another example of an operad, which we will encounter quite a lot in the following, we need to recall the definition of the braid group first.

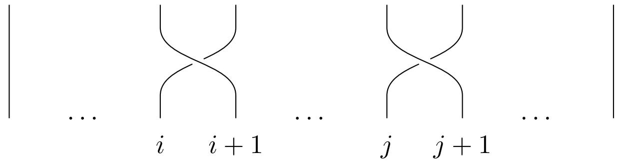
Definition 1.3. For each integer $n > 2$ we define the *braid group* on n strands as the group with $n-1$ generators $\sigma_1, \dots, \sigma_{n-1}$ satisfying the following relations:

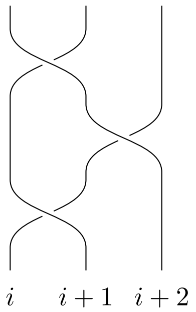
$$\begin{aligned} \sigma_i\sigma_j &= \sigma_j\sigma_i && \text{for } |i-j| > 1 \\ \sigma_i\sigma_{i+1}\sigma_i &= \sigma_{i+1}\sigma_i\sigma_{i+1} && \text{for } 1 \le i \le n-2 \end{aligned}$$

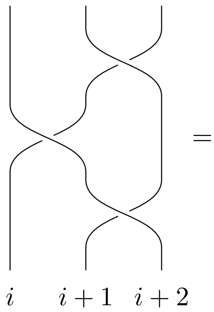
For $n = 2$ we define B_2 as the free group with one generator and let $B_0 = B_1 = \{1\}$ be the trivial group.

Remark 1.4. The name braid group B_n makes sense when we consider the following depiction of the group for $0 \le i, j \le n$ with $i + 1 < j$:

$\sigma_i =$ 

$\sigma_i \circ \sigma_j =$  $= \sigma_j \circ \sigma_i$

$\sigma_i \circ \sigma_{i+1} \circ \sigma_i =$ 

$=$ 

$= \sigma_{i+1} \circ \sigma_i \circ \sigma_{i+1}$

Further details on how this is defined formally can be found in [SC].

We state some facts about braid groups that are known and can for example be found in [YD].

Remark 1.5. 1. There is a projection, which we denote by $\pi\colon B_n \to S_n$, that adds the condition $\sigma_i^2 = \mathrm{Id}$.

2. The direct sum homomorphism

$$\bigoplus_S \colon S_{k_1} \times \cdots \times S_{k_n} \to S_{k_1+\cdots+k_n}$$

can be lifted to the braid groups:

$$\bigoplus_B \colon B_{k_1} \times \cdots \times B_{k_n} \to B_{k_1+\cdots+k_n}$$

such that $\bigoplus_B \circ \pi = (\pi)^k \circ \bigoplus_S$. Geometrically, $b_1 \oplus \cdots \oplus b_n \in B_{k_1+\ldots k_n}$ describes the braid that places $b_1, \ldots, b_n$ side by side from left to right.

3. For $n \geq 1$ and $k_1, \ldots, k_n \in \mathbb{N}$, the block permutation map

$$(-)_S(k_1, \ldots k_n)\colon S_n \to S_{k_1+\cdots+k+n}$$

can be lifted to the braid groups:

$$(-)_B(k_1, \ldots k_n)\colon B_n \to B_{k_1+\cdots+k_n}$$

such that $(-)_B(k_1, \ldots k_n) \circ \pi = \pi \circ (-)_S(k_1, \ldots k_n)$. Geometrically, for $b \in B_n$, the corresponding block braid $b(k_1, \ldots, k_n)$ is obtained by replacing the i-th strand by k_i many parallel strands for $1 \leq i \leq n$.

4. Let $\sigma, \rho \in B_n$ and $\tau_j \in B_{k_j}$ for $1 \leq j \leq n$. Then the following hold in $B_{k_1+\cdots+k_n}$:

$$(\sigma\rho)(k_1, \ldots, k_n) = \sigma(k_{\bar{\rho}^{-1}(1)}, \ldots, k_{\bar{\rho}^{-1}(n)}) \cdot \rho(k_1, \ldots, k_n)$$
$$\sigma(k_1, \ldots, k_n) \cdot (\tau_1 \oplus \cdots \oplus \tau_n) = (\tau_{\bar{\sigma}^{-1}(1)} \oplus \cdots \oplus \tau_{\bar{\sigma}^{-1}(n)} \cdot \sigma(k_1, \ldots, k_n)$$

where $\bar{\sigma} + \pi(\sigma)$ and $\bar{\rho} + \pi(\rho)$.

Example 1.6. Let $\mathcal{B}$ be the *braid operad* defined by:

$$\mathcal{B}(n) := B_n,$$

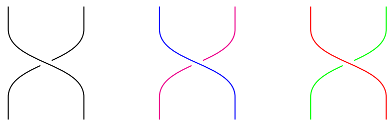
where B_n denotes the braid group on n strands. The composition morphism

$$\begin{aligned} \gamma\colon B_k \times B_{j_1} \times \cdots \times B_{j_k} &\to B_j \\ (\tau, (\sigma_i)_{1 \leq i \leq k}) &\mapsto \tau(\sigma_1 \oplus \cdots \oplus \sigma_k) \end{aligned}$$

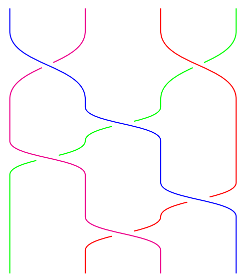
where $\tau(\sigma_1 \oplus \cdots \oplus \sigma_k)$ denotes the braid that τ-braids k many block braids $\sigma_1, \ldots, \sigma_k$ as if they where strands in B_k.
So for the compositon map

$$B_2 \times B_2 \times B_2 \to B_4$$

let us observe what happens with the following element in $B_2 \times B_2 \times B_2$:



After applying the composition map we obtain the following braiding:



The associativity and unitality conditions follow from the group conditions.

Definition 1.7. Let $\mathcal{P}, \mathcal{P}'$ be planar operads in a symmetric monoidal category $\mathcal{C}$. A *morphism of planar operads* $\beta\colon \mathcal{P} \to \mathcal{P}'$ consists of morphisms $\beta_n\colon P(n) \to \mathcal{P}'(n)$ for all $n \in \mathbb{N}$ such that the following diagrams commute:

$$\begin{tikzcd} \mathcal{P}(k) \otimes \mathcal{P}(j_1) \otimes \cdots \otimes \mathcal{P}(j_k) \arrow[r, "\beta_k \otimes \beta_{j_1} \otimes \cdots \otimes \beta_{j_k}"] \arrow[d, "\gamma"'] & \mathcal{P}'(k) \otimes \mathcal{P}'(j_1) \otimes \cdots \otimes \mathcal{P}'(j_k) \arrow[d, "\gamma'"] \\ \mathcal{P}(j) \arrow[r, "\beta_j"] & \mathcal{P}'(j) \end{tikzcd}$$

$$\begin{tikzcd} & k \arrow[dl, "\eta"'] \arrow[dr, "\eta'"] & \\ \mathcal{P}(1) \arrow[rr, "\beta_1"] & & \mathcal{P}'(1) \end{tikzcd}$$

Therefore, for any closed symmetric monoidal category $\mathcal{C}$ the planar operads and their morphisms form a category $\mathbf{PlanarOp}_{\mathcal{C}}$.

When we later discuss operads, we will seldom focus on planar operads but will work with symmetric operads instead.

Definition 1.8. A *(symmetric) operad* $\mathcal{O}$ in $\mathcal{C}$ is a planar operad as defined in 1.1 with an action of S_n on $\mathcal{O}(n)$ together with the following equivariance condition: Let $\sigma(j_1, \cdots, j_k) \in S_j$ denote the element that permutes k blocks of length j_l for $1 \leq l \leq k$. Then, the following diagram commutes:

$$\begin{tikzcd}
\mathcal{O}(k) \otimes \mathcal{O}(j_1) \otimes \cdots \otimes \mathcal{O}(j_k) \arrow[r, "\sigma\otimes\sigma^{-1}"] \arrow[d, "\gamma"'] & \mathcal{O}(k) \otimes \mathcal{O}(j_{\sigma(1)}) \otimes \cdots \otimes \mathcal{O}(j_{\sigma(k)}) \arrow[d, "\gamma"] \\
\mathcal{O}(j) \arrow[r, "\sigma(j_1\cdots j_k)"'] & O(j).
\end{tikzcd}$$

Furthermore, let $\tau_l \in S_{j_l}$ and $\tau_1 \oplus \cdots \oplus \tau_k$ denotes the sum of permutations on k blocks of length j_l for $1 \leq l \leq k$. Then the following diagram commutes:

$$\begin{tikzcd}
\mathcal{O}(k) \otimes \mathcal{O}(j_1) \otimes \cdots \otimes \mathcal{O}(j_k) \arrow[r, "\mathrm{Id}\otimes\tau_1\otimes\cdots\otimes\tau_k"] \arrow[d, "\gamma"'] & \mathcal{O}(k) \otimes \mathcal{O}(j_{\sigma(1)}) \otimes \cdots \otimes \mathcal{O}(j_{\sigma(k)}) \arrow[d, "\gamma"] \\
\mathcal{O}(j) \arrow[r, "\tau_1\oplus\cdots\oplus\tau_k"'] & \mathcal{O}(j).
\end{tikzcd}$$

As already mentioned we introduce operads as a tool to generally encode algebraic structure.

Definition 1.9. Let O be an operad in C. An *O-algebra* is an object A in C with maps

$$\theta\colon O(j) \otimes A^j \to A$$

for $j \in \mathbb{N}$ such that the following associativity, unitality and equivariant conditions hold:

- Associative:

$$\begin{tikzcd}
O(k) \otimes O(j_1) \otimes \cdots \otimes O(j_k) \otimes A^j \arrow[r, "\gamma\otimes\mathrm{Id}"] \arrow[dd, "s"'] & O(j) \otimes A^j \arrow[d, "\theta"] \\
 & A \\
O(k) \otimes (\bigotimes(O(j_l) \otimes A^{j_l})) \arrow[r, "\mathrm{Id}\otimes\theta^k"'] & O(k) \otimes A^k \arrow[u, "\theta"']
\end{tikzcd}$$

- Unital:

$$\begin{tikzcd}
\mathrm{I} \otimes A \arrow[r, "\cong"] \arrow[d, "\eta\otimes\mathrm{Id}"'] & A \\
O(1) \otimes A \arrow[ur, "\theta"'] &
\end{tikzcd}$$

- Equivariant: For all $j \in \mathbb{N}$ and $\sigma \in S_j$:

$$\begin{tikzcd}
O(j) \otimes A^j \arrow[rr, "\sigma\otimes\sigma^{-1}"] \arrow[dr, "\theta"'] & & O(j) \otimes A^j \arrow[dl, "\theta"] \\
 & A &
\end{tikzcd}$$

Definition 1.10. Let O be an operad in $\mathcal{C}$, A an O-algebra. We call an object M in $\mathcal{C}$ with maps

$$\omega\colon O(j) \otimes A^{j-1} \otimes M \to M$$

an *A-module* if it satisfies the following associativity, unitality and equivariance conditions:

- Associative: For all $\sum j_l = j$ in $\mathbb{N}$ the following diagrams commute:

$$
\begin{tikzcd}
O(k) \otimes O(j_1) \otimes \cdots \otimes O(j_k) \otimes A^{j-1} \otimes M \arrow[r, "\gamma \otimes \mathrm{Id}"] \arrow[dd, "s"'] & O(j) \otimes A^{j-1} \otimes M \arrow[d, "\omega"] \\
 & M \\
O(k) \otimes \left(\bigotimes_{l=1}^{k-1} O(j_l) \otimes A^{j_l} \right) \otimes O(j_k) \otimes A^{j_k - 1} \otimes M \arrow[r, "\mathrm{Id} \otimes \theta^{k-1} \otimes \omega"'] & O(k) \otimes A^{k-1} \otimes M \arrow[u, "\omega"']
\end{tikzcd}
$$

- Unital:

$$
\begin{tikzcd}
\mathrm{I} \otimes M \arrow[r, "\cong"] \arrow[d, "\eta \otimes \mathrm{Id}"'] & M \\
O(1) \otimes A^0 \otimes M \arrow[ur, "\omega"'] &
\end{tikzcd}
$$

- Equivariant: For all $j \in \mathbb{N}$ and $\sigma \in S_{j-1} \subset S_j$, the following diagram commutes:

$$
\begin{tikzcd}
O(j) \otimes A^{j-1} \otimes M \arrow[rr, "\sigma \otimes \sigma^{-1} \otimes \mathrm{Id}"] \arrow[dr, "\omega"'] & & O(j) \otimes A^{j-1} \otimes M \arrow[dl, "\omega"] \\
 & M &
\end{tikzcd}
$$

Remark 1.11. We can of course also define algebras over planar operads by dropping the equivariance condition as there is no symmetric group action. So, algebras over the operad $\mathrm{Ass}^{\otimes}$ which we defined in 1.2 are indeed monoids since the corresponding squares commute due to canonical isomorphisms:

$$
\begin{tikzcd}
\mathrm{Ass}^{\otimes}(2) \otimes \mathrm{Ass}^{\otimes}(2) \otimes \mathrm{Ass}^{\otimes}(1) \otimes A^3 \arrow[r, "\cong"] \arrow[dd, "\cong"'] & \mathrm{Ass}^{\otimes}(3) \otimes A^3 \arrow[d, "\theta"] \\
 & A \\
\mathrm{Ass}^{\otimes}(2) \otimes (\mathrm{Ass}^{\otimes}(2) \otimes A^2) \otimes (\mathrm{Ass}^{\otimes}(1) \otimes A)) \arrow[r, "\mathrm{Id} \otimes \theta^2"'] & \mathrm{Ass}^{\otimes}(2) \otimes A^2 \arrow[u, "\theta"']
\end{tikzcd}
$$

$$
\begin{tikzcd}
\mathrm{Ass}^{\otimes}(2) \otimes \mathrm{Ass}^{\otimes}(1) \otimes \mathrm{Ass}^{\otimes}(2) \otimes A^3 \arrow[r, "\cong"] \arrow[dd, "\cong"'] & \mathrm{Ass}^{\otimes}(3) \otimes A^3 \arrow[d, "\theta"] \\
 & A \\
\mathrm{Ass}^{\otimes}(2) \otimes (\mathrm{Ass}^{\otimes}(1) \otimes A) \otimes (\mathrm{Ass}^{\otimes}(2) \otimes A^2)) \arrow[r, "\mathrm{Id} \otimes \theta^2"'] & \mathrm{Ass}^{\otimes}(2) \otimes A^2 \arrow[u, "\theta"']
\end{tikzcd}
$$

Examples 1.12. 1. A natural question is whether the associative planar operad and the associative operad in a category $\mathcal{C}$ coincide. For $\mathcal{C} = \mathbf{Set}$, we can equip the

$$
\mathrm{Ass}^{\otimes \text{planar}}_{\mathbf{Set}}(n) = *
$$

with the trivial action of S_n. Yet this appears to be too restrictive: The equivariance condition implies that algebras over this operad are not just monoids but commutative monoids. To distinguish operations after acting by S_n, we can think of a more suited candidate: $\mathrm{Ass}^{\otimes}(n) = S_n$ for all $n \in \mathbb{N}$ where S_n acts on $\mathrm{Ass}^{\otimes}(n)$ by composition. The composition map is defined by considering a sum of permutations and then permuting k blocks of length j_i for $1 \leq i \leq k$:

$$\begin{aligned}\gamma\colon\ \mathrm{Ass}^{\otimes}(k) \times \mathrm{Ass}^{\otimes}(j_1) \times \cdots \times \mathrm{Ass}^{\otimes}(j_k) &\to \mathrm{Ass}^{\otimes}(k)\\ (\sigma, \tau_1, \ldots, \tau_k) &\mapsto \sigma(\tau_1 \oplus \cdots \oplus \tau_k)\end{aligned}$$

We might also refer to this operad as the symmetry operad $\mathcal{S}$ later on.

2. The operad $\mathrm{Comm}^{\otimes}_{\mathbf{Set}}(n) = *$ with the trivial action of S_n is an interesting operad nevertheless as it is the *commutative operad* in the category **Set**. The commutative operad in a category $\mathcal{C}$ is the operad such that its algebras are commutative monoids in $\mathcal{C}$.

3. The *little n-disk operad* D_n is an important example of an operad in the category of topological spaces **Top**. For all $n \in \mathbb{N}$ we will define $\mathrm{D}_n(k)$ as a topological space representing the space of k disjoint little disks inside the unit disk in the plane. Let $n \in \mathbb{N}$ and

$$\mathrm{D}^n = \left\{(r_1, \ldots, r_n) \in \mathbb{R}^n \colon r_1^2 + \cdots + r_n^2 \leq 1\right\} \subset \mathbb{R}^n$$

is the closed unit n-disc. A little n-disc is defined as the embedding $f\colon\ \mathrm{D}^n \to \mathrm{D}^n$ of the following form:

$$f(r_1, \ldots, r_n) \coloneqq (c_1, \ldots, c_n) + \lambda(r_1, \ldots, r_n)$$

for $(c_1, \ldots, c_n) \in \mathrm{D}^n$ and $\lambda \in \mathbb{R}$ such that

$$0 < \lambda \leq 1 - \sum_{i=1}^{n} c_i^2.$$

We denote the interior of a little disc by $\mathrm{int}(f)$ and define for $k > 0$:

$$\mathrm{D}_n(k) \coloneqq \left\{ f_1 \sqcup \cdots \sqcup f_n \in C\left(\bigsqcup_{i=1}^{k} \mathrm{D}^n, \mathrm{D}^n\right) \ :\ \mathrm{int}(f_i) \cap \mathrm{int}(f_j) = \emptyset \text{ for all } i \neq j \right\}$$

We equip $C\left(\bigsqcup_{i=1}^{k} \mathrm{D}^n, \mathrm{D}^n\right)$ with the compact-open topology and $\mathrm{D}_n(k)$ with the subspace topology. $\mathrm{D}_n(0)$ is the one point space. There is an operad structure to be defined on

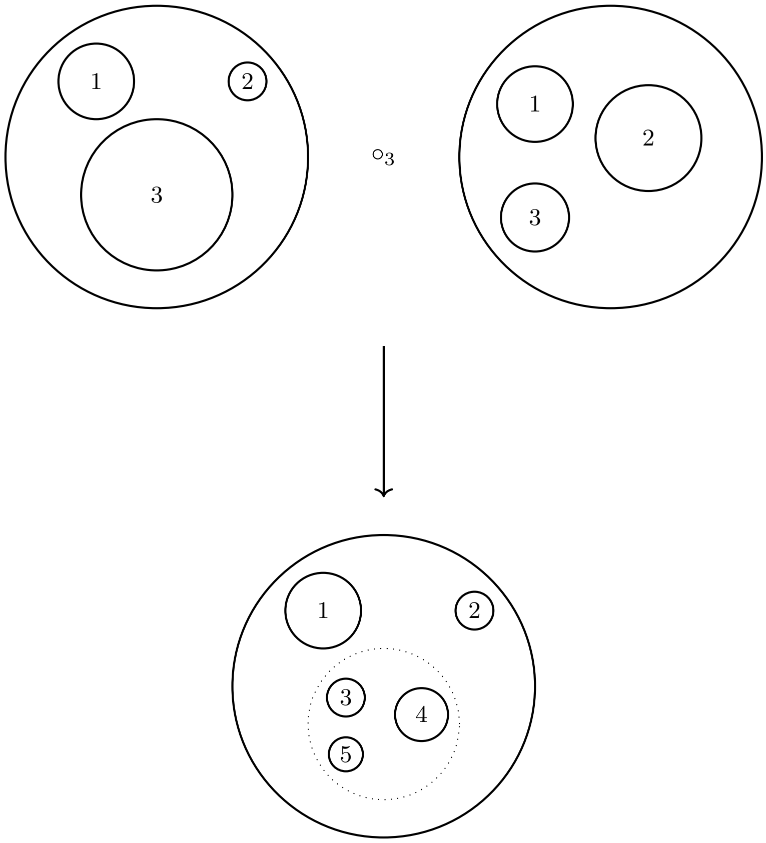
$$\mathrm{D}_n = \{\mathrm{D}_n(k)\}_{k \geq 0}\,.$$

The construction of the composition map

$$\gamma : \mathrm{D}_n(k) \times \mathrm{D}_n(l_1) \times \cdots \times \mathrm{D}_n(l_k) \to \mathrm{D}_n(l_1 + \cdots + l_k)$$

is geometrically interpreted as:

- Start with a configuration in $\mathrm{D}_n(k)$, which gives k disks inside the unit disk.
- Each of the k disks is treated as a new "unit disk" in which a configuration from $\mathrm{D}_n(l_1), \mathrm{D}_n(l_2), \ldots, \mathrm{D}_n(l_k)$ can be inserted.
- The configurations from $\mathrm{D}_n(l_1), \mathrm{D}_n(l_2), \ldots, \mathrm{D}_n(l_k)$ are scaled down to fit inside the respective disks from the configuration in $\mathrm{D}_n(k)$, and they are inserted.
- After placing all configurations inside the k disks, the seams are erased, yielding a new configuration in $\mathrm{D}_n(n_1 + \cdots + n_k)$.



Formally, this means that for $(f, (g_i)_{1\leq i\leq k}) \in \mathrm{D}_n(l_1) \times \mathrm{D}_n(l_2) \times \cdots \times \mathrm{D}_n(l_k)$ we define $\gamma((f, (g_i)_{1\leq i\leq k}))$ via the following commuting diagram:

$$\begin{tikzcd}
\bigsqcup_{j=1}^{l} \mathrm{D}^n \arrow[r, "\gamma(f,(g_i)_{1\leq i\leq k})"] \arrow[d, "\cong"] & \mathrm{D}^n \\
\bigsqcup_{i=1}^{k} \left(\bigsqcup_{p=1}^{l_k} \mathrm{D}^n\right) \arrow[r, "\bigsqcup g_i"] & \bigsqcup_{i=1}^{k} \mathrm{D}^n \arrow[u, "f"]
\end{tikzcd}$$

The unit is the identity configuration, where the single disk remains unchanged, i.e. the identity map $\mathrm{Id}_{\mathrm{D}^n} \in C\left(\mathrm{D}^n, \mathrm{D}^n\right)$. This satisfies the unitality condition for operads by definition, where inserting a single disk into another disk does not alter any configuration: $\gamma((f, (\mathrm{Id})_{1\leq i\leq k})) = f$ and $\gamma(\mathrm{Id}, g) = g$.

The composition map satisfies the associativity condition by definition. In other words inserting disks into one another is associative.

The disks are labeled, and there is an action of the symmetric group S_k on $\mathrm{D}_n(k)$, which permutes the labels of the disks: For $f_1 \sqcup \cdots \sqcup f_k \in \mathrm{D}_n(k)$ and $\sigma \in S_k$:

$$\sigma(f_1 \sqcup \cdots \sqcup f_k) \coloneqq f_{\sigma(1)} \sqcup \cdots \sqcup f_{\sigma(k)}$$

The composition map γ respects the action of S_k: For any $\sigma \in S_k$, $\tau_i \in S_{l_i}$ and

$(f,(g_i)_{1\leq i\leq k}) \in \mathrm{D}_n(k)\times \mathrm{D}_n(l_1)\times\cdots\times \mathrm{D}_n(l_k)$ we have by definition

$$\begin{aligned}\sigma\circ\gamma((f,(g_i)_{1\leq i\leq k})) &= \sigma\left(\bigsqcup_{i=1}^{k} f_i\circ g_i\right)\\ &= \bigsqcup_{i=1}^{k} f_{\sigma(i)}\circ g_{\sigma(i)}\\ &= \gamma\left((f_{\sigma(1)}\sqcup\cdots\sqcup f_{\sigma(k)}, g_{\sigma(1)},\ldots,g_{\sigma(k)}\right)\\ &= \gamma\circ(\sigma\times\sigma^{-1})((f,(g_i)_{1\leq i\leq k}))\end{aligned}$$

and

$$\begin{aligned}(\tau_1\oplus\cdots\oplus\tau_k)\circ\gamma((f,(g_i)_{1\leq i\leq k})) &= (\tau_1\oplus\cdots\oplus\tau_k)\left(\bigsqcup_{i=1}^{k} f_i\circ g_i\right)\\ &= \bigsqcup_{i=1}^{k} f_i\circ\tau_i(g_i)\\ &= \gamma\left((f,\tau_1(g_1),\ldots,\tau_k(g_k)\right)\\ &= \gamma\circ(\tau_1\times\cdots\times\tau_k)((f,(g_i)_{1\leq i\leq k})).\end{aligned}$$

1.2 Colored Operads

Following [LJ], the goal of this section is to introduce colored operads in such a way that we can define colored infinite operads later on. As we will see, a symmetric monoidal category is an example of a colored operad. Thus, we will focus on this special case first. We will show that a symmetric monoidal category can be defined in a purely functorial way. After that we will then follow this approach and define the notion of a colored operad in a similar way.
Let us start by introducing some terminology.

Definition 1.13. Let $P\colon \mathcal{C}\to\mathcal{D}$ be a functor of arbitrary categories and $f\colon x\to y$ a morphism in $\mathcal{C}$. We call f *cartesian* if for any morphism $g\colon z\to y$ in $\mathcal{C}$ and $\omega\colon P(z)\to P(x)$ in $\mathcal{D}$ with $P(f)\circ\omega = P(g)$, there exists a unique morphism $\omega'\colon z\to x$ in $\mathcal{C}$ such that $g = f\circ\omega'$ and $P(\omega') = \omega$. In diagrams we have:

$$\begin{tikzcd} z \arrow[r, "g"] \arrow[d, dotted, "\exists!"'] & y & & P(z) \arrow[r] \arrow[d] & P(y) \\ x \arrow[ur, "f"'] & & {} \arrow[r, "P"] & P(x) \arrow[ur] & \end{tikzcd}$$

And vice versa, we call f *cocartesian* if we can flip arrows of the above case, i.e. for any morphism $g\colon y\to z$ in $\mathcal{C}$ and $\omega\colon P(x)\to P(z)$ in $\mathcal{D}$ with $\omega\circ P(f) = P(g)$, there is a unique morphism $\omega'\colon x\to z$ such that $P(\omega') = \omega$ and $\omega'\circ f = g$. In a diagram this means the following:

$$\begin{tikzcd} y \arrow[r] \arrow[dr, "f"'] & z & {} \arrow[r, "P"] & {} & P(y) \arrow[r] \arrow[dr] & P(z) \\ & x \arrow[u, dotted, "\exists!"'] & & & & P(x) \arrow[u] \end{tikzcd}$$

Note that equivalently we can define a cocartesian morphism by requiring a bijection:

$$\mathrm{Hom}_{\mathcal{C}}(y,z)\times_{\mathrm{Hom}_{\mathcal{D}}(P(y),P(z))}\mathrm{Hom}_{\mathcal{D}}(P(x),P(z)) \cong \mathrm{Hom}_{\mathcal{C}}(x,z)$$

where $\mathrm{Hom}_{\mathcal{C}}(y,z) \times_{\mathrm{Hom}_{\mathcal{D}}(P(y),P(z))} \mathrm{Hom}_{\mathcal{D}}(P(x),P(z))$ is the pullback of applying the functor P to $\mathrm{Hom}_{\mathcal{C}}(y,z)$ and precomposing with $P(f)\colon P(y) \to P(x)$. So, for the bijection we send a pair (g,ω) to the unique morphism ω' and vice versa we send a morphism $h\colon x \to z$ to the pair $(h \circ f, P(h))$.

Definition 1.14. Let $P\colon \mathcal{C} \to \mathcal{D}$ be a functor. P is called a *(Grothendieck) fibration* if for every object y in $\mathcal{C}$ and morphisms $f'\colon x' \to P(y)$ there exists a cartesian morphism $f\colon x \to y$ such that $f' = P(f)$. We call f a cartesian lifting of f' and f' a cleavage. Dually we define the notion of a *(Grothendieck) cofibration*, i.e. P is a cofibration if for every object x in $\mathcal{C}$ and morphism $f'\colon P(x) \to y'$ there exists a cocartesian morphism $f\colon x \to y$ such that $P(f) = f'$.

Definition 1.15. Let $\mathcal{C}$ be a symmetric monoidal category. We define a new category $\mathcal{C}^{\otimes}$ in the following way:

- Objects of $\mathcal{C}^{\otimes}$ are finite sequences $[C_1,\dots,C_n]$ where C_i is an object in $\mathcal{C}$ for all i. It is possible that the sequence is empty.
- A morphismn $f\colon [C_1,\dots,C_n] \to [C'_1,\dots,C'_m]$ in $\mathcal{C}^{\otimes}$ consists of a subset $S_f \subset \{1,\dots,n\}$, a map of finite sets $\alpha_f\colon S_f \to \{1,\dots,m\}$ and a family of morphisms

$$\left\{ f_j\colon \bigotimes_{\alpha_f(i)=j} C_i \to C'_j \right\}_{1\le j\le m}$$

  in $\mathcal{C}$.
- Composition is defined as follows: Let

$$f\colon [C_1,\dots,C_n] \to [C'_1,\dots,C'_m] \quad \text{and} \quad g\colon [C'_1,\dots,C'_m] \to [C''_1,\dots,C''_k]$$

  be morphisms in $\mathcal{C}^{\otimes}$, then

$$g \circ f\colon [C_1,\dots,C_n] \to [C''_1,\dots,C''_k]$$

  is defined via $S_{g\circ f} = \alpha_f^{-1}(S_g)$, $\alpha_{g\circ f} = \alpha_g \circ \alpha_f|_{S_{g\circ f}}$ and for all $1 \le l \le k$ we have:

$$\bigotimes_{\alpha_{g\circ f}(i)=l} C_i \cong \bigotimes_{\alpha_g(j)=k} \bigotimes_{\alpha_g(i)=j} C_i \to \bigotimes_{\alpha_g(j)=k} C'_j \to C''_l.$$

Let I be a finite set. We write $I_* = I \coprod *$.

Definition 1.16. We define $\mathcal{F}\mathrm{in}_*$, the *category of finite pointed sets*, as follows:

- For each $n \in \mathbb{N}$ we have $\langle n\rangle^\circ := \{1,\dots,n\}$. We define the objects of $\mathcal{F}\mathrm{in}_*$ as $\langle n\rangle := \langle n\rangle^\circ_*$.
- A morphism $\alpha\colon \langle n\rangle \to \langle m\rangle$ is a map of finite sets where $\alpha(*) = *$

It is clear that this defines a category. Further note that this category is equivalent to the category of finite sets with a distinguished point since for any finite set I we have an isomorphism $I_* \cong \langle n\rangle$, where n is the cardinality of I. For all pairs of $1 \le i \le n$ we write $\rho^i\colon \langle n\rangle \to \langle 1\rangle$ for the morphism in $\mathcal{F}\mathrm{in}_*$ which is defined by

$$\rho^i(j) = \begin{cases} 1 & i = j \\ * & \text{else.} \end{cases}$$

For any symmetric monoidal category $\mathcal{C}$ there exists a forgetful functor

$$P\colon \mathcal{C}^{\otimes} \to \mathcal{F}\mathrm{in}_*$$

which sends a sequence $[C_1, \dots, C_n]$ to $\langle n \rangle$ and a morphism $f\colon [C_1, \dots, C_n] \to [C'_1, \dots, C'_m]$ to $\alpha\colon \langle n \rangle \to \langle m \rangle$ with

$$\alpha(x) = \begin{cases} \alpha_f(x) & x \in S_f \\ * & \text{else.} \end{cases}$$

Remark 1.17. Let $\mathcal{C}_0$ denote the trivial category of one object and one morphism. We claim that the forgetful functor $P\colon \mathcal{C}_0^\otimes \to \mathcal{F}\mathrm{in}_*$ is an isomorphism of categories. This holds, since for all morphisms in $\mathcal{C}_0^\otimes$ the family of morphisms f_j in $\mathcal{C}_0$ are trivial, in other words there is nothing to forget. Hence, $\mathcal{F}\mathrm{in}_*$ is obtained as the simplest example of the construction of a $\mathcal{C}^\otimes$ category.
Furthermore, we note that the forgetful functor $P\colon \mathcal{C}^\otimes \to \mathcal{F}\mathrm{in}_*$ can be obtained from a unique symmetric monoidal functor $F\colon \mathcal{C} \to \mathcal{C}_0$ using the functoriality of F.

Remark 1.18. For every object $\langle n \rangle$ we denote the fiber category by $\mathcal{C}_{\langle n \rangle}$, which is defined to be the pullback of the following diagram

$$\begin{tikzcd} \mathcal{C}^\otimes_{\langle n \rangle} \arrow[r, dashed] \arrow[d, dashed] & \mathcal{C}^\otimes \arrow[d, "P"] \\ \{\langle n \rangle\} \arrow[r, hook] & \mathcal{F}\mathrm{in}_* \end{tikzcd}$$

where $\{\langle n \rangle\}$ is interpreted as the trivial one object category. In other words, $\mathcal{C}_{\langle n \rangle}$ consists of all collections of objects of length n and all morphisms f such that $\alpha_f = \mathrm{Id}_{\langle n \rangle}$.

We are now ready to state two very important properties of the functor $P\colon \mathcal{C}^\otimes \to \mathcal{F}\mathrm{in}_*$, which as it turns out is all we need to know.

Proposition 1.19. *(M1) The forgetful functor $P\colon \mathcal{C}^\otimes \to \mathcal{F}in_*$ is a cofibration.*

(M2) We have that $\mathcal{C} \cong \mathcal{C}^\otimes_{\langle 1 \rangle}$ and more precisely we obtain that $\mathcal{C}^\otimes_{\langle n \rangle} \cong \mathcal{C}^n$ where $\mathcal{C}^n$ denotes the n-fold product of copies of $\mathcal{C}$.

Proof. (M1) Let $[C_1, \dots, C_n]$ be an object in $\mathcal{C}^\otimes$ and $f'\colon \langle n \rangle \to \langle m \rangle$ a morphism in $\mathcal{F}\mathrm{in}_*$. We need to construct a cocartesian morphism $f\colon [C_1, \dots, C_n] \to [C'_1, \dots, C'_m]$ such that $P(f) = f'$. We define $\alpha_f\colon S_f \to \langle m \rangle^\circ$ as $\alpha_f := f'|_{S_f}$ where $S_f := f'^{-1}(\langle m \rangle^\circ)$. Further we define a selection of objects in $\mathcal{C}$ for $1 \leq j \leq m$:

$$C'_j := \begin{cases} \bigotimes_{\alpha_f(i)=j} C_i & j \in \alpha_f(S_f) \\ \mathrm{I} & \text{else} \end{cases}$$

For $f_j := \mathrm{Id}_{C'_j}$ we have then constructed a well defined morphism

$$f\colon [C_1, \dots, C_n] \to [C'_1, \dots, C'_m]$$

such that $P(f) = f'$. We are left to show that f is indeed cocartesian. Therefore, let $g\colon [C_1, \dots, C_n] \to [C''_1, \dots, C''_l]$ be a morphism in $\mathcal{C}^\otimes$ and $\omega'\colon \langle m \rangle \to \langle l \rangle$ a morphism in $\mathcal{F}\mathrm{in}_*$ such that $P(g) = \omega' \circ P(f)$. We need to find a unique morphism

$$\omega\colon [C'_1, \dots, C'_m] \to [C''_1, \dots, C''_l]$$

such that $g = \omega \circ f$ and $P(\omega) = \omega'$. We are forced to define $\alpha_\omega := \omega'|_{S_\omega}$ where

$$S_\omega = \omega'^{-1}(\langle l \rangle^\circ).$$

By construction we have $\alpha_g = \alpha_\omega \circ \alpha_f$ and by definition of C'_j we are forced to define for all $1 \leq j \leq l$:

$$\omega_j\colon \bigotimes_{\alpha_\omega(k)=j} C'_k \cong \bigotimes_{\alpha_\omega(k)=j} \bigotimes_{\alpha_f(i)=k} C_i \cong \bigotimes_{\alpha_g(k)=j} C_k \xrightarrow{g_j} C''_j$$

Uniqueness of ω follows from $\mathcal{C}$ being a symmetric monoidal category. Hence, f is cocartesian.

(M2) Let $F\colon \mathcal{C}^{\otimes}_{\langle 1\rangle} \to \mathcal{C}$ be a functor that sends $[C]$ to C. Let $f\colon [C] \to [C']$ be a morphism in $\mathcal{C}^{\otimes}_{\langle 1\rangle}$, then $\alpha_f = \mathrm{Id}_{\{1\}}$ and there is just one morphism $f_1\colon C \to C'$ corresponding to f. So, we define $F(f) = f_1$. It follows by definition that F is fully faithful and essentially surjective, i.e. an equivalence of categories.

First, we will show that for any morphism $\langle n\rangle \to \langle m\rangle$ (M1) induces a functor of fiber categories $A\colon \mathcal{C}^{\otimes}_{\langle n\rangle} \to \mathcal{C}^{\otimes}_{\langle m\rangle}$: Let $\alpha\colon \langle n\rangle \to \langle m\rangle$ be a morphism in $\mathcal{F}\mathrm{in}_*$ and $[C_1, \dots, C_n]$ an object in $\mathcal{C}^{\otimes}_{\langle n\rangle}$. By (M1), we can find a cocartesian morphism

$$f_\alpha\colon [C_1, \dots, C_n] \to [C'_1, \dots, C'_m].$$

Therefore, we can define

$$A([C_1, \dots, C_n]) \coloneqq [C'_1, \dots, C'_m].$$

To see that the choice of $[C'_1, \dots, C'_m]$ is unique up to isomorphism, let

$$f'_\alpha\colon [C_1, \dots, C_n] \to [\tilde{C}'_1, \dots, \tilde{C}'_m]$$

be a different choice of cocartesian morphism for $[C_1, \dots, C_n]$ and α. Then there are morphisms

$$g\colon [C'_1, \dots, C'_m] \to [\tilde{C}'_1, \dots, \tilde{C}'_m] \quad \text{and} \quad g'\colon [\tilde{C}'_1, \dots, \tilde{C}'_m] \to [C'_1, \dots, C'_m]$$

corresponding to the cocartesian property of f_α and f'_α respectively. Because of uniqueness we obtain $g \circ g' = \mathrm{Id}_{[\tilde{C}'_1, \dots, \tilde{C}'_m]}$ and $g' \circ g = \mathrm{Id}_{[C'_1, \dots, C'_m]}$. Therefore, the choice of a cocartesian morphism is unique up to isomorphism.

Let $f\colon [C_1, \dots, C_n] \to [\bar{C}_1, \dots, \bar{C}_n]$ be a morphism in $\mathcal{C}^{\otimes}_{\langle n\rangle}$. We obtain a unique morphism $f'\colon [C'_1, \dots, C'_m] \to [\bar{C}'_1, \dots, \bar{C}'_m]$ from the fact that f_α is cocartesian.

$$\begin{tikzcd}
{[C_1, \dots, C_n]} \arrow[r, "f_\alpha"] \arrow[d, "f"] & {[C'_1, \dots, C'_m]} \arrow[d, dashed, "f'"] & & \langle n\rangle \arrow[r, "\alpha"] \arrow[d, "\mathrm{Id}"] & \langle m\rangle \arrow[d, "\mathrm{Id}"] \\
{[\bar{C}_1, \dots, \bar{C}_n]} \arrow[r, "\bar{f}_\alpha"] & {[\bar{C}'_1, \dots, \bar{C}'_m]} & {} \arrow[r, phantom, "\xrightarrow{P}"] & \langle n\rangle \arrow[r, "\alpha"] & \langle m\rangle
\end{tikzcd}$$

Hence, we can define $A(f) \coloneqq f'$. Due to uniqueness of f', A preserves composition and maps identities to identities.

Let $P^i\colon \mathcal{C}^{\otimes}_{\langle n\rangle} \to \mathcal{C}^{\otimes}_{\langle 1\rangle} \cong \mathcal{C}$ be the functor corresponding to $\rho^i\colon \langle n\rangle \to \langle 1\rangle$ which we defined previously for all $1 \le i \le n$. Then from the universal property of the product, it follows that

$$\prod_{1\le i\le n} P^i\colon \mathcal{C}^{\otimes}_{\langle n\rangle} \to \mathcal{C}^n$$

is an equivalence of categories.

□

Note that in the following we might refer to the constructed functor $\mathcal{C}^{\otimes}_{\langle n\rangle} \to \mathcal{C}^{\otimes}_{\langle m\rangle}$ originated from $\alpha\colon \langle n\rangle \to \langle m\rangle$ also by α.

We are now well prepared to see that the symmetric monoidal structure on $\mathcal{C}$ is decoded in $\mathcal{C}^{\otimes}$ and its forgetful functor $P\colon \mathcal{C}^{\otimes} \to \mathcal{F}\mathrm{in}_*$ satisfying the above conditions. In other words, for any category $\mathcal{D}$ together with a functor $P\colon \mathcal{D} \to \mathcal{F}\mathrm{in}_*$ satisfying (M1) and (M2), we can construct a symmetric monoidal category $\mathcal{C}$ such that $\mathcal{D} \cong \mathcal{C}^{\otimes}$ and P is the usual forgetful functor. The right candidate for $\mathcal{C}$ is clearly $\mathcal{D}_{\langle 1\rangle}$.

Proposition 1.20. *There is a bijection*

$$\{\textit{symmetric monoidal categories}\} \cong \{\textit{cofibrations } \mathcal{D} \to \mathcal{F}in_* \textit{ satisfying (M2)}\}$$

Proof. Let $\mathcal{D}$ be a category and $P\colon \mathcal{D} \to \mathcal{F}\mathrm{in}_*$ a functor satisfying (M1) and (M2). We will equip $\mathcal{C} := \mathcal{D}_{\langle 1\rangle}$ with a symmetric monoidal structure and see that there is an equivalence of categories $\mathcal{C}^{\otimes} \cong \mathcal{D}$.

1. For the map $\alpha\colon \langle 2\rangle \to \langle 1\rangle$ with $\alpha^{-1}(1) = \{1,2\}$ we obtain the following functor using (M2):
$$\bigotimes\colon \mathcal{C} \times \mathcal{C} \cong \mathcal{D}_{\langle 2\rangle} \xrightarrow{\alpha} \mathcal{C}$$

2. There is a unique morphism $\eta\colon \langle 0\rangle \to \langle 1\rangle$ in $\mathcal{F}\mathrm{in}_*$ defining a functor $\mathcal{D}_{\langle 0\rangle} \to \mathcal{C}$ which gives us a unique object I in $\mathcal{C}$. This object I will serve as the tensor unit which follows from the following commuting diagram:
$$\begin{tikzcd}
\mathcal{D}_{\langle 0\rangle} \times \mathcal{D}_{\langle 1\rangle} \arrow[r, "\eta\times\mathrm{Id}"] & \mathcal{D}_{\langle 1\rangle} \times \mathcal{D}_{\langle 1\rangle} \arrow[r, "\cong"] & \mathcal{D}_{\langle 2\rangle} \arrow[dr, "\otimes"] & \\
\mathcal{C} \arrow[u, "\cong"'] \arrow[d, "\cong"] & & & \mathcal{D}_{\langle 1\rangle} \\
\mathcal{D}_{\langle 1\rangle} \times \mathcal{D}_{\langle 0\rangle} \arrow[r, "\mathrm{Id}\times\eta"] & \mathcal{D}_{\langle 1\rangle} \times \mathcal{D}_{\langle 1\rangle} \arrow[r, "\cong"] & \mathcal{D}_{\langle 2\rangle} \arrow[ur, "\otimes"] &
\end{tikzcd}$$

3. Furthermore, we need a symmetric braiding. This we obtain from the morphism
$$\begin{aligned} \beta\colon \langle 2\rangle &\to \langle 2\rangle \\ 1 &\mapsto 2 \\ 2 &\mapsto 1. \end{aligned}$$

   We note that $\alpha\circ\beta = \alpha$ and $\rho^i\beta = \rho^{\beta(i)}$. So, we obtain the following commuting diagramm:
$$\begin{tikzcd}
D_{\langle 1\rangle} \times D_{\langle 1\rangle} & D_{\langle 2\rangle} \arrow[l, "\rho^1\times\rho^2"'] \arrow[r, "\alpha"] \arrow[d, "\beta"] & D_{\langle 1\rangle} \\
& D_{\langle 2\rangle} \arrow[ul, "\rho^2\times\rho^1"] \arrow[ur, "\alpha"'] &
\end{tikzcd}$$

   Hence, $A \otimes B \cong B \otimes A$ holds.

4. The associator corresponds to the following maps in $\mathcal{F}\mathrm{in}_*$:
$$\begin{aligned} \tau_i^n\colon \langle n\rangle &\to \langle n-1\rangle \\ j &\mapsto \begin{cases} j & 1 \le j \le i \\ j-1 & i < j \le n \\ * & j = * \end{cases} \end{aligned}$$

   So, the following commuting diagram
$$\begin{tikzcd}
\langle 3\rangle \arrow[r, "\tau_1^3"] \arrow[d, "\tau_2^3"'] & \langle 2\rangle \arrow[d, "\tau_1^2"] \\
\langle 2\rangle \arrow[r, "\tau_1^2"] & \langle 1\rangle
\end{tikzcd}$$

translates to $(A \otimes B) \otimes C \cong A \otimes (B \otimes C)$. We need to see the pentagon identities:

$$\begin{tikzcd}
 & (A \otimes B) \otimes (C \otimes D) \arrow[dl] \arrow[dr] & \\
A \otimes (B \otimes (C \otimes D)) & & ((A \otimes B) \otimes C) \otimes D \arrow[d] \\
A \otimes ((B \otimes C) \otimes D) \arrow[u] & & (A \otimes (B \otimes C)) \otimes D \arrow[ll]
\end{tikzcd}$$

The commutativity of the following diagram in $\mathcal{F}\mathrm{in}_*$ translates to the pentagon identities:

$$\begin{tikzcd}
 & \langle 3 \rangle \arrow[r, "\tau_2^3"] \arrow[dr, "\tau_1^3"] & \langle 2 \rangle \arrow[dr, "\tau_1^2"] & \\
\langle 4 \rangle \arrow[ur, "\tau_1^4"] \arrow[r, "\tau_2^4"] \arrow[dr, "\tau_3^4"] & \langle 3 \rangle \arrow[r, "\tau_1^3"] \arrow[dr, "\tau_2^3"] & \langle 2 \rangle \arrow[r, "\tau_1^2"] & \langle 1 \rangle \\
 & \langle 3 \rangle \arrow[r, "\tau_2^3"] & \langle 2 \rangle \arrow[ur, "\tau_1^2"] &
\end{tikzcd}$$

$$\begin{array}{lcl}
\tau_1^2 \circ \tau_2^3 \circ \tau_1^4 & \rightsquigarrow & (A \otimes B) \otimes (C \otimes D) \\
\tau_1^2 \circ \tau_1^3 \circ \tau_1^4 & \rightsquigarrow & ((A \otimes B) \otimes C) \otimes D \\
\tau_1^2 \circ \tau_1^3 \circ \tau_2^4 & \rightsquigarrow & (A \otimes (B \otimes C)) \otimes D \\
\tau_1^2 \circ \tau_2^3 \circ \tau_2^4 & \rightsquigarrow & A \otimes ((B \otimes C) \otimes D) \\
\tau_1^2 \circ \tau_2^3 \circ \tau_3^4 & \rightsquigarrow & A \otimes (B \otimes (C \otimes D))
\end{array}$$

□

Remark 1.21. With some adjustments this construction can also be done for a monoidal category $\mathcal{C}$. The category $\mathcal{C}^{\otimes}$ is defined analoguously to the symmetric case. But we choose ordered indexing sets $[n]^{\circ} := \{0 < \cdots < n\}$ and for a morphism $f\colon [C_1, \ldots, C_n] \to [C'_1, \ldots, C'_m]$ we require that α_f preserves the order. Further, we can define the *pointed simplex category* Δ_* as follows:

- For each $n \in \mathbb{N}$ we define an object in Δ_* as $[n] := [n]^{\circ} \sqcup \{*\}$.
- The single pointed set $\{*\}$ is an object in Δ_*. There are morphisms $\{*\} \hookrightarrow [n]$ for all n.
- A morphism $\alpha\colon [n] \to [m]$ is a order preserving map on $\alpha^{-1}([m])$ and $\alpha(*) = *$.

Note that for all pairs $0 \leq i \leq n$, we have $\rho^i\colon [n] \to [0]$ in Δ_* with

$$\rho^i(j) = \begin{cases} 0 & i = j \\ * & \text{else.} \end{cases}$$

Similarly to the symmetric case, we obtain a forgetful functor:

$$P\colon \mathcal{C}^{\otimes} \to \Delta_*$$

This functor is a cofibration. The proof in 1.19 applies. Note that for the proof the fact that morphisms in Δ_* preserve order is essential, otherwise we could not define

$$\omega_j\colon \bigotimes_{\alpha_\omega(k)=j} C'_k \cong \bigotimes_{\alpha_\omega(k)=j} \bigotimes_{\alpha_f(i)=k} C_i \cong \bigotimes_{\alpha_g(k)=j} C_k \xrightarrow{g_j} C''_j$$

in a monoidal category $\mathcal{C}$. Further, the (M2) property applies as it is a result of the (M1) property and the fact that ρ^i is a morphism in Δ_*. It then also follows that there is a bijection

$$\{\text{monoidal categories}\} \cong \{\text{cofibration } \mathcal{D} \to \Delta_* \text{ satisfying (M2)}\}$$

The proof for this is analoguously to 1.20 as all morphisms in $\mathcal{F}\mathrm{in}_*$ that induce a monoidal structure on $\mathcal{C}$ are order preserving and can be considered in Δ_*.

We will now introduce the notion of a colored operad and will move on to see that, similarly to describing a symmetric monoidal structure via a functor satisfying (M1) an (M2), there is also a functor satisfying similar properties that encode the datum of a colored operad.

Definition 1.22. A *colored operad* $\mathcal{O}$ consists of the following data:

(i) A collection $\{X, Y, Z; \dots\}$ which we call objects or colors of the operad.

(ii) For every finite collection of objects $\{X_i\}_{i\in I}$ and object Y in $\mathcal{O}$, there is a set $\mathrm{Mul}_{\mathcal{O}}(\{X_i\}_{i\in I}, Y)$ called the set of morphisms from $\{X_i\}_{i\in I}$ to Y.

(iii) For any finite map $I \to J$ with fibers denoted by I_j for all $j \in J$, collections of objects $\{X_i\}_{i\in I}$ and $\{Y_j\}_{j\in J}$ and object Z, there is a composition map

$$\prod_{j\in J} \mathrm{Mul}_{\mathcal{O}}(\{X_i\}_{i\in I_j}, Y_j) \times \mathrm{Mul}_{\mathcal{O}}(\{Y_j\}_{j\in J}, Z) \to \mathrm{Mul}_{\mathcal{O}}(\{X_i\}_{i\in I}, Z)$$

(iv) There is a collection of morphisms $\{\mathrm{Id}_X \in \mathrm{Mul}_{\mathcal{O}}(\{X\}, X)\}_{X\in\mathcal{O}}$ being a left and right unit in the composition, i.e. the following diagrams commute:

$$\begin{tikzcd}
\mathrm{Mul}_{\mathcal{P}}(X_I, Y) \arrow[r, "\cong"] \arrow[d, "\mathrm{Id}"] & \mathrm{Mul}_{\mathcal{P}}(X_I, Y) \times \{\mathrm{Id}_Y\} \arrow[d, hook] \\
\mathrm{Mul}_{\mathcal{P}}(X_I, Y) & \mathrm{Mul}_{\mathcal{P}}(X_I, Y) \times \mathrm{Mul}_{\mathcal{P}}((Y), Y) \arrow[l, hook']
\end{tikzcd}$$

$$\begin{tikzcd}
\mathrm{Mul}_{\mathcal{P}}(X_I, Y) \arrow[r, "\cong"] \arrow[d, "\mathrm{Id}"] & (\prod_{i\in I}\{\mathrm{Id}_{X_i}\}) \times \mathrm{Mul}_{\mathcal{P}}(X_I, Y) \arrow[d, hook] \\
\mathrm{Mul}_{\mathcal{P}}(X_I, Y) & (\prod_{i\in I} \mathrm{Mul}_{\mathcal{P}}((X_i), X_i)) \times \mathrm{Mul}_{\mathcal{P}}(X_I, Y) \arrow[l, hook']
\end{tikzcd}$$

(v) For every sequence of finite maps $I \to J \to K$, collections of objects $X_I = \{X_i\}_{i\in I}$, $Y_J = \{Y_j\}_{j\in J}$, $W_K = \{W_k\}_{k\in K}$ and object $Z \in \mathcal{O}$ the following diagram commutes:

$$\begin{tikzcd}
\prod_{j\in J} \mathrm{Mul}_{\mathcal{O}}(X_I, Y_j) \times \prod_{k\in K} \mathrm{Mul}_{\mathcal{O}}(Y_J, W_k) \times \mathrm{Mul}_{\mathcal{O}}(W_K, Z) \arrow[dr, "\mathrm{Id}\times\gamma"] \arrow[dd, "(\gamma\circ s)\times\mathrm{Id}"'] & \\
& \prod_{j\in J} \mathrm{Mul}_{\mathcal{O}}(X_{I_j}, Y_j) \times \mathrm{Mul}_{\mathcal{O}}(Y_J, Z) \arrow[dd] \\
\prod_{k\in K} \mathrm{Mul}_{\mathcal{O}}(X_{I_k}, W_k) \times \mathrm{Mul}_{\mathcal{O}}(W_K, Z) \arrow[dr] & \\
& \mathrm{Mul}_{\mathcal{O}}(X_I, Z)
\end{tikzcd}$$

Remark 1.23. When we drop the symmetric structure we obtain a definition of a colored operad what we call colored planar operad. In other word we consider the indexing sets I in 1.22 to be finite ordered sets and maps between indexing sets have to be order preserving.

Remark 1.24. We note that there is a forgetful functor from the categories of colored operads to the category of small categories. Hence, for every colored operad $\mathcal{O}$ there is an underlying category which we also denote by $\mathcal{O}$. Its objects are the colors of $\mathcal{O}$ and for $X, Y \in \mathcal{O}$ we define the morphisms

$$\mathrm{Hom}(X, Y) := \mathrm{Mul}_{\mathcal{O}}(\{X\}, Y).$$

Composition and identity hold since they hold in $\mathcal{O}$. Thus, a colored operad can be viewed as a category with additional structure. Since we have some sort of collection of multilinear maps, some also refer to colored operads by multicategories.

Examples 1.25.

1. A colored operad with one color is a symmetric operad in **Set** as defined in 1.8. Let $\mathcal{O}$ be a colored operad with one color X. We will construct a symmetric operad $\mathcal{P}$ from $\mathcal{O}$. Since $\mathcal{O}$ is a colored operad with only one color, we define the family of objects $\mathcal{P}(n)$ for $n \in \mathbb{N}$ as:

   $$\mathcal{P}(n) := \mathrm{Mul}_{\mathcal{O}}(\{X\}_{1 \leq i \leq n}, X).$$

   The composition maps from $\mathcal{O}$ for natural number indexing then defines the composition maps of $\mathcal{P}$

   $$\gamma : \mathcal{P}(k) \times \mathcal{P}(j_1) \times \cdots \times \mathcal{P}(j_k) \to \mathcal{P}(j)$$

   where $j = \sum_{i=1}^{k} j_i$ and $\eta \colon \{*\} \to \mathcal{P}(1)$ is given by Id_X. Unitality and associativity follow the corresponding conditions in $\mathcal{O}$. However, in a colored operad (even with one color), the set of morphisms depends not only on n, the cardinality of the indexing set, but also on the specific choice of finite indexing set. Thus, for two sets N_1 and N_2 with the same number of elements, you could potentially have different morphism sets:

   $$\mathrm{Mul}_{\mathcal{O}}(\{X_i\}_{i \in N_1}, Y) \quad \text{and} \quad \mathrm{Mul}_{\mathcal{O}}(\{X_i\}_{i \in N_2}, Y).$$

   This raises the question of how the structures of a colored operad and an operad reconcile. The reason these two different setups are still equivalent lies in the symmetric group action on the set of morphisms. The permuation of the inputs ensures that any specific labeling of the inputs (as represented by the different finite sets N_1 and N_2) can be transformed into each other via the action of S_n. Hence, although the morphisms in the colored operad are indexed by specific sets, they behave the same way up to this permutation action. Therefore, the group action effectively collapses the dependence on the specific finite sets. So, while the morphism sets in a colored operad are technically different for different finite sets, the permutation symmetry makes these differences irrelevant, aligning the structure with that of an operad.

2. A colored planar operad with one color is a planar operad in **Set** as defined in 1.1. Let $\mathcal{P}$ be a colored planar operad $\mathcal{P}$. We set for $n \in \mathbb{N}$:

   $$\mathcal{P}(n) := \mathrm{Mul}_{\mathcal{P}}((X)_{1 \leq i \leq n}, X)$$

   The composition map and operadic unit we obtain similarly as in the example for a one colored operad. However, as the indexing maps need to be order preserving we will not obtain a permutation of labels as there is only one bijective order preserving map between finite ordered sets.

Examples 1.26.

1. Let $\mathcal{C}$ be a symmetric monoidal category. By setting for a finite family of objects $\{X_i\}_{i \in I}$ and object Y in $\mathcal{C}$

   $$\mathrm{Mul}_{\mathcal{C}}(\{X_i\}_{i \in I}, Y) := \mathrm{Hom}(\otimes_{i \in I} X_i, Y)$$

   we can define a colored operad $\mathcal{C}$. The composition is given by

   $$\prod_{j \in J} \mathrm{Hom}(\otimes_{i \in I_j} X_i, Y_j) \times \mathrm{Hom}(\otimes_{j \in J} Y_j, Z) \to \mathrm{Hom}(\otimes_{i \in I} X_i, Z)$$

   $$((f_j)_{j \in J}, g) \mapsto g \circ (\otimes_{j \in J} f_j)$$

with $\{\mathrm{Id}_X\}_{X \in \mathcal{C}}$ being the unit and associativity holds since for $(f_j \colon \bigotimes_{i \in I_j} X_i \to Y_j)_{j \in J}$, $(g_k \colon \bigotimes_{j \in J_k} Y_j \to W_k)_{k \in K}$, $h \colon \bigotimes_{k \in K} W_k \to Z$ we have

$$h \circ \bigotimes_{k \in K} \left(g_k \circ \bigotimes_{j \in J_k} f_j \right) = h \circ \bigotimes_{k \in K} g_k \circ \bigotimes_{j \in J} f_j.$$

Further, we can reconstruct the symmetric monoidal structure of $\mathcal{C}$ from this colored operad: Let $F_{X,Y} \colon \mathcal{C} \to \mathbf{Set}$ be the functor sending an object Z to $\mathrm{Mul}_{\mathcal{C}}(\{X, Y\}, Z)$ then by definition and from Yonedas lemma follows that F is uniquely (up to isomorphism) represented by $X \otimes Y$. Let $f \colon X \to X'$ and $g \colon Y \to Y'$ be morphisms in $\mathcal{C}$ and

$$\mathrm{Mul}_{\mathcal{C}}(\{X\}, X') \times \mathrm{Mul}_{\mathcal{C}}(\{Y\}, Y') \times \mathrm{Mul}_{\mathcal{C}}(\{X', Y'\}, X' \otimes Y') \xrightarrow{\gamma} \mathrm{Mul}_{\mathcal{C}}(\{X, Y\}, X' \otimes Y')$$

the compositon map. Then, we define $f \otimes g \coloneqq \gamma((f, g, \mathrm{Id}))$. The tensor unit I is the representing object of the functor sending an object Z in $\mathcal{C}$ to $\mathrm{Mul}_{\mathcal{C}}(\emptyset, Z)$. This example is even clearer when we have the functorial definition of a colored operad later on. As it will be a generalization of 1.19 and 1.20 illustrates how to reconstruct the symmetric monoidal category.

2. Analogously, a monoidal category $\mathcal{C}$ is an example of a colored planar operad.

Definition 1.27. Let $\mathcal{O}$ be a colored operad. We define a category $\mathcal{O}^{\otimes}$ in the following way:

(i) The objects of $\mathcal{O}^{\otimes}$ are finite sequences of colors of $\mathcal{O}$.

(ii) Let $\{X_i\}_{1 \le i \le m}$ and $\{Y_j\}_{1 \le j \le n}$ be objects in $\mathcal{O}^{\otimes}$. A morphism

$$\phi \colon \{X_i\}_{1 \le i \le m} \to \{Y_j\}_{1 \le j \le n}$$

is given by a map $\alpha \colon \langle m \rangle \to \langle n \rangle$ in $\mathcal{F}\mathrm{in}_*$ together with a family of morphisms

$$\{\phi_j \in \mathrm{Mul}_{\mathcal{O}}(\{X_i\}_{i \in \alpha^{-1}(j)}, Y_j)\}_{1 \le j \le n}$$

in $\mathcal{O}$.

(iii) Composition of morphisms is determined by the composition in $\mathcal{F}\mathrm{in}_*$ and $\mathcal{O}$, i.e. for morphisms $\phi \colon X \to Y$ and $\phi' \colon Y \to Z$ in $\mathcal{O}^{\otimes}$ with $\alpha \colon \langle n \rangle \to \langle m \rangle$ and $\alpha' \colon \langle m \rangle \to \langle k \rangle$, we have $\bar{\alpha} = \alpha' \circ \alpha$ and

$$\{(\phi' \circ \phi)_l \in \mathrm{Mul}_{\mathcal{O}}(\{X_i\}_{i \in \bar{\alpha}^{-1}(l)}, Z_l)\}_{1 \le l \le k}$$

where $(\phi' \circ \phi)_l$ is applying the composition map

$$\gamma \colon \prod_{j \in \alpha'^{-1}(l)} \mathrm{Mul}_{\mathcal{O}}(\{X_i\}_{i \in \alpha^{-1}(j)}, Y_j) \times \mathrm{Mul}_{\mathcal{O}}(\{Y_i\}_{i \in \alpha'^{-1}(l)}, Z_l) \to \mathrm{Mul}_{\mathcal{O}}(\{X_i\}_{i \in \bar{\alpha}^{-1}(l)}, Z_l)$$

Remark 1.28. For a planar colored operad $\mathcal{P}$, we can similarly define a category $\mathcal{P}^{\otimes}$ by requiring $\alpha \colon [m] \to [n]$ to be morphism in the simplex category with a distinguished point Δ_*.

Let $\mathcal{O}$ be a colored operad. Then similarly as in 1.15 we have a forgetful functor

$$\pi \colon \mathcal{O}^{\otimes} \to \mathcal{F}\mathrm{in}_*$$

from which we can reconstruct the underlying colored operad O (up to canonical equivalence): First, note that in general π does not have to be a cofibration. For an arbitrary morphism $\alpha \colon \langle n \rangle \to \langle m \rangle$ and a collection $\{X_i\}_{i \in \langle n \rangle}$ there is no canonical choice for a collection $\{X'_j\}_{j \in \langle m \rangle}$ with morphism $f \colon \{X_i\}_{i \in \langle n \rangle} \to \{X'_j\}_{j \in \langle m \rangle}$ such that $\pi(f) = \alpha$ as in the proof of 1.19. However, there is a canonical choice if α is a bijection on $\alpha^{-1}(\langle m \rangle^{\circ})$. Then for all $j \in \langle m \rangle$ there is exactly

one $i_j \in \langle n \rangle$ with $\alpha(i_j) = j$. Hence, we define $X'_j := X_{i_j}$ and for all $j \in J$ we have $f_j = \mathrm{Id}_{X_{i_j}}$. This defines a morphism $f\colon \{X_i\}_{i \in I} \to \{X'_j\}_{j \in J}$.
It is left to show that f is in fact cocartesian. Let $g\colon \{X_i\}_{i \in I} \to \{X''_k\}_{k \in \langle l \rangle}$ be a morphism in $\mathcal{O}^{\otimes}$ and $\omega\colon \langle m \rangle \to \langle l \rangle$ a morphisn in $\mathcal{F}\mathrm{in}_*$ such that $\pi(g) = \omega \circ \pi(f)$. Due to the fact of $\pi(f)$ being a bijection on the restriction, we observe that ω is determined by $\pi(g)$. It follows that

$$\mathrm{Mul}(\{X'_j\}_{j \in \omega^{-1}(k)}, X''_k) = \mathrm{Mul}(\{X_{a_j}\}_{j \in \omega^{-1}(k)}, X''_k) = \mathrm{Mul}(\{X_i\}_{i \in \pi(g)^{-1}(k)}, X''_k)$$

and we can define for all $k \in \langle l \rangle$ $h_k = g_k$. So, we have constructed a morphism

$$h\colon \{X'_j\}_{j \in \langle m \rangle} \to \{X''_k\}_{k \in \langle l \rangle}$$

such that $h \circ f = g$ and $\pi(h) = \omega$. The underlying category of $\mathcal{O}$ is the fiber category $\mathcal{O}_{\langle 1 \rangle}$ as the objects are 1-collections of colors and a morphism from X to Y is defined by $\mathrm{Id}_{\langle 1 \rangle}$ with $\phi \in \mathrm{Mul}_{\mathcal{O}}(\{X\}, Y)$. Furthermore, the equivalence $\mathcal{O}_{\langle n \rangle} \cong \mathcal{O}^n$ holds as the weaker (M1) property still suffices to admit the proof given in 1.19.
Further, let $\{X_i\}_{i \in \langle n \rangle}$ be a collection of colors, $\vec{X}$ denoting the corresponding object in $\mathcal{O}^{\otimes}_{\langle n \rangle}$ and $Y \in \mathcal{O}$. The set $\mathrm{Mul}_{\mathcal{O}}(\{X_i\}_{i \in \langle n \rangle}, Y)$ can be identified with the set of morphisms $f\colon \vec{X} \to Y$ in $\mathcal{O}^{\otimes}$ such that $\pi(f)$ being the morphism with $\pi(f)^{-1}(*) = \{*\}$. In addition, the composition law for morphisms in the colored operad $\mathcal{O}$ can be reconstructed from the composition law in $\mathcal{O}^{\otimes}$.

Remark 1.29. Let $\mathcal{P}$ be a colored planar operad. Then analogously there is also a forgetful functor

$$\pi\colon \mathcal{P}^{\otimes} \to \Delta_*.$$

which satisfies the weaker (M1) and (M2) property.

This gives us an alternative definition of a colored operad. Instead of seeing a colored operad as a categorical structure $\mathcal{O}$ equipped with an extensive notion of morphisms, we can now view colored operads as a category $\mathcal{O}^{\otimes}$ equipped with a forgetful functor $\pi\colon \mathcal{O}^{\otimes} \to \mathcal{F}\mathrm{in}_*$ satisfying conditions such as $\mathcal{O}^{\otimes}_{\langle n \rangle} \cong (\mathcal{O}^{\otimes}_{\langle 1 \rangle})^n$ and $\mathcal{O} = \mathcal{O}^{\otimes}_{\langle 1 \rangle}$ inherit the structure of a colored operad. Even though the conditions π has to satisfy seem tedious, the second definition is phrased entirely in the language of category theory and can therefore be generalized to a ∞-categorical notion of a colored operad.

Definition 1.30. A morphism $f\colon \langle n \rangle \to \langle m \rangle$ is called *inert* if for every $i \in \langle m \rangle^{\circ}$ the inverse image $f^{-1}(i)$ has exactly one element.

Definition 1.31. An ∞-*operad* is a functor $p\colon \mathcal{O}^{\otimes} \to \mathcal{N}(\mathcal{F}\mathrm{in}_*)$ between ∞-categories satisfying the following conditions:

(i) For every inert morphism $f\colon \langle n \rangle \to \langle m \rangle$ in $\mathcal{N}(\mathcal{F}\mathrm{in}_*)$ and object $C \in \mathcal{O}^{\otimes}_{\langle m \rangle}$ there exists a p-cocartesian morphism $f\colon C \to C'$ in $\mathcal{O}^{\otimes}$ lifting f. In particular f induces a functor $f_!\colon \mathcal{O}^{\otimes}_{\langle n \rangle} \to \mathcal{O}^{\otimes}_{\langle m \rangle}$.

(ii) Let $C \in \mathcal{O}^{\otimes}_{\langle m \rangle}$ and $C' \in \mathcal{O}^{\otimes}_{\langle n \rangle}$ be objects, $f : \langle m \rangle \to \langle n \rangle$ be a morphism in $\mathcal{F}\mathrm{in}_*$ and let $\mathrm{Map}^f_{\mathcal{O}^{\otimes}}(C, C')$ be the union of connected components of $\mathrm{Map}_{\mathcal{O}^{\otimes}}(C, C')$ which lie over f. Choose p-cocartesian morphisms $C' \to C'_i$ lying over the inert morphism $\rho^i\colon \langle n \rangle \to \langle 1 \rangle$ for $1 \le i \le n$. Then the induced map

$$\mathrm{Map}^f_{\mathcal{O}^{\otimes}}(C, C') \to \prod_{1 \le i \le n} \mathrm{Map}^{\rho^i \circ f}_{\mathcal{O}^{\otimes}}(C, C'_i)$$

is a homotopy equivalence

(iii) For every finite collection of objects $C_1, \ldots, C_n \in \mathcal{O}^{\otimes}_{\langle 1 \rangle}$, there exists an object $C \in \mathcal{O}^{\otimes}_{\langle n \rangle}$ and a collection of p-cocartesian morphism $C \to C_i$ covering $\rho^i\colon \langle n \rangle \to \langle 1 \rangle$.

2 Braided Operads

Braided operads were first mentioned by Fiedorowicz in his paper [FZ] on construction of E_n operads. Let A be an operad in the category of compactly generated Hausdorff spaces. The first theorem states:

Theorem 2.1 (Recognition principles for E_n operads)**.** *A is an E_2 operad iff $A(k)$ is connected and the collection of universal covering spaces $\{\tilde{A}(k)\}_{k\in\mathbb{N}}$ forms a B_∞ operad, i.e.:*

(1) For all $k \in \mathbb{N}$ $\tilde{A}(k)$ is contractible.

(2) The braid group B_k acts freely on $\tilde{A}(k)$ for each $k \in \mathbb{N}$.

In the proof, it is stated that "the definition of operads can be reformulated using actions by braid groups in place of actions by symmetric groups". As braided operads are a useful tool, e.g. recognizing E_n operads, we want to gain a deeper understanding of braided operads. Further, we have already seen that colored operads correspond to fibrations over finite pointed sets $\mathcal{F}\mathrm{in}_*$ as well as colored planar operads correspond to fibrations over Δ_*. We are hoping to find a suitable category such that colored braided operads correspond to fibrations over this category. It might be interesting to try to define braided operads in braided monoidal categories rather than symmetric monoidal categories potentially yielding interesting examples. However, it is not straight forward how to do this as shuffling objects would not be a canonical isomorphism. So for now, we will stay in a symmetric monoidal category $\mathcal{C}$.

Definition 2.2. A *braided operad* $\mathcal{O}$ in $\mathcal{C}$ is a planar operad as defined in 1.1 with an action of B_n on $\mathcal{O}(n)$ for all $n \in \mathbb{N}$ together with the following equivariance condition: For $\sigma \in B_k$, let $\sigma(j_1, \cdots, j_k) \in B_j$ denote the element that σ-braids k blocks with each block having j_l strands for $1 \leq l \leq k$. Then the following diagram commutes:

$$\begin{tikzcd}
\mathcal{O}(k) \otimes \mathcal{O}(j_1) \otimes \cdots \otimes \mathcal{O}(j_k) \arrow[r, "\sigma\otimes\bar{\sigma}^{-1}"] \arrow[d, "\gamma"'] & \mathcal{O}(k) \otimes \mathcal{O}(j_{\bar{\sigma}(1)}) \otimes \cdots \otimes \mathcal{O}(j_{\bar{\sigma}(k)}) \arrow[d, "\gamma"] \\
\mathcal{O}(j) \arrow[r, "\sigma(j_1\cdots j_k)"'] & O(j),
\end{tikzcd}$$

where $\bar{\sigma}$ denotes the underlying permutation of σ. Furthermore, let $\tau_l \in B_{j_l}$ for $1 \leq l \leq k$ and $\tau_1 \oplus \cdots \oplus \tau_k$ denotes the sum of braidings on k blocks with each block having j_l strands. Then the following diagram commutes

$$\begin{tikzcd}
\mathcal{O}(k) \otimes \mathcal{O}(j_1) \otimes \cdots \otimes \mathcal{O}(j_k) \arrow[r, "\mathrm{Id}\otimes\tau_1\otimes\cdots\otimes\tau_k"] \arrow[d, "\gamma"'] & \mathcal{O}(k) \otimes \mathcal{O}(j_{\sigma(1)}) \otimes \cdots \otimes \mathcal{O}(j_{\sigma(k)}) \arrow[d, "\gamma"] \\
\mathcal{O}(j) \arrow[r, "\tau_1\oplus\cdots\oplus\tau_k"'] & \mathcal{O}(j)
\end{tikzcd}$$

Remark 2.3. Similarily to symmetric operads, we can define algebras and modules over braided operads by changing the equivariance condition:
Let $\mathcal{O}$ be a braided operad and A an algebra over the underlying planar operad of $\mathcal{O}$. For all $j \in \mathbb{N}$, $\sigma \in B_j$ and $\bar{\sigma} \in S_j$ being the underlying permutation of σ:

$$\begin{tikzcd}
O(j) \otimes A^j \arrow[rr, "\sigma\otimes\bar{\sigma}^{-1}"] \arrow[dr, "\theta"'] & & O(j) \otimes A^j \arrow[dl, "\theta"] \\
& A &
\end{tikzcd}$$

Then, we call A an algebra over the braided operad $\mathcal{O}$.
Further, let A be an algebra over the braided operad $\mathcal{O}$ and M an object in $\mathcal{C}$ with

$$\omega \colon O(j) \otimes A^{j-1} \otimes M \to M$$

satisfying associativity and unitality conditions from 1.10. Then we call M an A-module if the following equvariance condition holds for all $j \in \mathbb{N}$, $\sigma \in B_{j-1} \subset B_j$ and $\bar{\sigma} \in S_{j-1}$ being the underlying permutation of σ:

$$\begin{tikzcd}
O(j) \otimes A^{j-1} \otimes M \arrow[rr, "\sigma\otimes\bar{\sigma}^{-1}\otimes \mathrm{Id}"] \arrow[dr, "\omega"'] & & O(j) \otimes A^{j-1} \otimes M \arrow[dl, "\omega"] \\
& M &
\end{tikzcd}$$

Example 2.4. 1. Again, let us consider the operad in **Set** defined by the singleton set in each degree $n \in \mathbb{N}$. The braid group acts trivially. When we consider algebras over this operad we will again obtain commutative algebras.

2. We have already encountered an example of planar operad in 1.6 which can be lifted to a braided operad—the braid operad $\mathcal{B}$. The braid group B_n acts by composition. This group action is equivariant.

2.1 Action Operad

For this section, we remain in the symmetric monoidal category of small sets equipped with the direct product as the monoidal product. We will introduce an action operad that acts on planar operads and we will see how this relates to braided operads following [YD].

Definition 2.5. An *action operad* is a tuple

$$(\{G(n)\}_{n\in\mathbb{N}}, \gamma^G, 1^G, \omega)$$

satisfying the following conditions:

(i) For all $n \in \mathbb{N}$, $G(n)$ is a group with unit Id_n.

(ii) There is a group homomorphism $\omega\colon G(n) \to S_n$ for all n which we call augmentation. For each $g \in G(n)$ we call $\omega(g)$ the underlying permutation of g.

(iii) $(\{G(n)\}_{n\in\mathbb{N}}, \gamma^G, 1^G)$ is a one-colored planar operad in **Set** which we denote by G. It fulfills the following conditions:

(a) The augmentation
$$\omega\colon G \to \mathcal{S}$$
is a morphism of planar operads in **Set** where $\mathcal{S}$ denotes the symmetry operad from 1.12.

(b) For $n \geq 1$, let $\sigma, \rho \in G(n)$ and for $k_j \in \mathbb{N}$, $1 \leq j \leq n$ let $\tau_j, \tau'_j \in G(k_j)$. Then the following equation holds in $G(k)$ for $\sum_{1\leq j\leq n} k_j = k$:

$$\gamma^G(\sigma\rho, \tau_1\tau'_1, \ldots, \tau_n\tau'_n) = \gamma^G(\sigma, \tau_{\bar{\rho}^{-1}(1)}, \ldots, \tau_{\bar{\rho}^{-1}(n)}) \cdot \gamma^G(\rho, \tau'_1, \ldots, \tau'_n)$$

where $\bar{\rho} = \omega(\rho) \in S_n$.

We call the condition (iii)(b) the action operad equation.

Examples 2.6. 1. The *planar group operad* P is the action operad where $P(n) = \{*\}$ is the trivial group for all $n \in \mathbb{N}$.

2. The symmetry operad $\mathcal{S}$ from 1.12 equipped with the identity augmentation forms an action operad.

3. Another example which we have already seen is the braid operad which we equip with the projection to the symmetric group as the augmentation.

Proposition 2.7. *The braid operad $\mathcal{B}$ from 1.6 equipped with the projection*

$$\pi\colon \mathcal{B} \to \mathcal{S}$$

forms an action operad

$$B = (\{B_n\}_{n\in\mathbb{N}}, \gamma, \mathrm{Id}_1 \in \mathcal{B}_1, \pi)$$

Proof. We have already seen that $\mathcal{B}$ forms a one-colored planar operad in **Set**. Hence, it remains to show that π is a morphism of planar operads. We note that $\pi(\mathrm{Id}_1)$ is the trivial permutation and π preserves the operadic composition as it perserves the group multiplication. The action operad equation follows from 1.5. □

2.2 Outlook

We want to define braided operads as fibrations over a suitable category. A first intuition led us to the assumption that colored braided operads might fiber over the crossed simplicial braid group as defined in [DK15].

Definition 2.8. The *crossed simplicial braid group* is a category ΔB equipped with an embedding $\iota : \Delta \to \Delta B$ such that:

1. the functor ι is bijective on objects,
2. any morphism $u : \iota[m] \to \iota[n]$ in ΔB can be uniquely expressed as a composition $\iota(\phi) \circ b$ where $\phi : [m] \to [n]$ is a morphism in Δ and $b \in B_m$ is acting on $\iota[m]$ in ΔB.

Note that we sometimes refer to $\iota[m]$ by writing $[m]$ instead.

Remark 2.9. To understand composition in the category ΔB let us consider two morphisms

$$f\colon [m] \to [n] \quad \text{and} \quad g\colon [n] \to [l]$$

with the factorization $f = \phi_f \circ b_f$ and $g = \phi_g \circ b_g$. We define for every $j \in [n]$ an f-pencil p_j as a collection $[j, i_1, \dots i_s]$ with $\phi_f^{-1}(j) = \{i_1 < \cdots < i_s\}$. We call s the width of the pencil. Note that we say it is an empty pencil p_j if $\phi_f^{-1}(j) = \emptyset$.
Then we obtain a braid on m strands in the following way: First, we consider $p_1, \dots, p_n$ as n strand such that we can braid by b_g. Then, we forget the empty pencil strands; in other words we restrict the action of b_g to the subset $J \subset [n]$ such that p_j is not empty for all $j \in J$. However, for notation reason and without loss of generality, we assume that there are no empty pencils. We obtain a braiding on m strands if we consider the action of b_g on the pencils as a block braiding on m strand. Each pencil corresponds to a block having the length of the pencils width. The blocks are ordered by $[n]$.
Geometrically this can be thought of as braiding the pencils. Then, as the order within the pencil remains unchanged, we can cut a pencil $[j, j_1, \dots, j_s]$ into its s underlying strands. The resulting braid corresponds to $b_g(s_1, \dots, s_n) \circ b_f$ where s_j is the width of the pencil p_j for $j \in [n]$. Thus, we define

$$g \circ f := \phi_g \circ (b_g(s_1, \dots, s_n) \circ b_f).$$

We can define a new colored operad with the crossed simplicial braid group as its indexing sets. However, this causes some issues. A braided monoidal category is supposed to be an example of the new colored operad, meaning we require a morphism in ΔB, inducing the braiding in the braided monoidal category. Thus, the following diagram would have to commute in ΔB:

$$\begin{tikzcd} {[1]} \arrow[r, "\tau_0^1"] \arrow[d, "\sigma_0"'] & {[0]} \\ {[1]} \arrow[ur, "\tau_0^1"'] & \end{tikzcd}$$

which is not the case.
The next step will be to investigate whether we can identify further relations to define a suitable category related to ΔB.

add potential fix and explain the contradiction of braid action on cartesian product

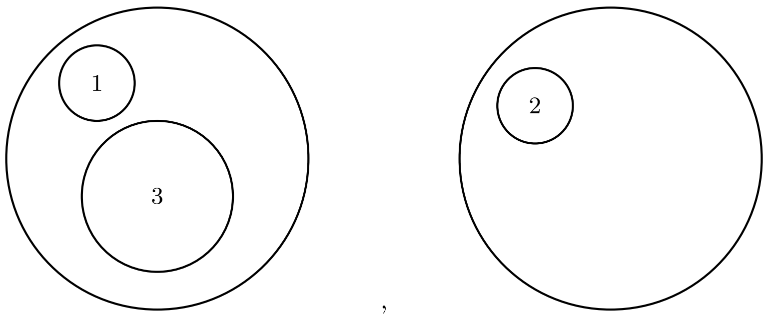
2.3 Braided monoidal structure

In the following we will construct a braided monoidal structure obtained from a cofibration over $\mathrm{E}_2^{\otimes}$. To do so we will need to choose paths in $\mathrm{E}_2^{\otimes}$ on various occations. We will refrain from specifying the paths in detail but rather describe the paths enough to understand from which homotopy class we choose a path.

Remark 2.10. Let us recall the definition of $\mathrm{E}_2^{\otimes}$. For each $n \in \mathbb{N}$ we have an object $\langle n \rangle$. A mophism $\langle n \rangle \to \langle m \rangle$ in $\mathrm{E}_2^{\otimes}$ consists of a morphism $\alpha\colon \langle n \rangle \to \langle m \rangle$ in $\mathcal{F}\mathrm{in}_*$ and a collection of configurations $\{c_j \in \mathrm{E}_2(|\alpha^{-1}(j)|)\}_{1 \leq j \leq m}$ where each configuration is labeled by $\alpha^{-1}(j))$. So for example, let

$$\begin{aligned} \alpha\colon \langle 3 \rangle &\to \langle 2 \rangle \\ 1 &\mapsto 1 \\ 2 &\mapsto 2 \\ 3 &\mapsto 1 \end{aligned}$$

and



,

We denote the trivial configuration with label x by $\mathrm{Id}^x \in \mathrm{E}_2(1)$.

Definition 2.11. Let $\mathcal{C}^{\otimes}$ be a category with a Grothendieck cofibration

$$\mathcal{C}^{\otimes} \to \mathrm{E}_2^{\otimes}$$

satisfying the (M2) condition: For $\mathcal{C} \coloneqq \mathcal{C}^{\otimes}_{\langle 1 \rangle}$, we have $\mathcal{C}^{\otimes}_{\langle n \rangle} \cong \mathcal{C}^n$. Then we call $\mathcal{C}^{\otimes}$ with π a E_2-operad.

Let $\pi\colon \mathcal{C}^{\otimes} \to \mathrm{E}_2^{\otimes}$ be a E_2-operad. We denote $\mathcal{C} \coloneqq \mathcal{C}^{\otimes}_{\langle 1 \rangle}$. In the following section we aim to define a braided monoidal structure on $\mathcal{C}$.
Note that for a diagram in $\mathrm{E}_2^{\otimes}$

$$\begin{tikzcd} \langle n \rangle \arrow[r, "c_1"] \arrow[dr, "c_2"'] & \langle 1 \rangle \arrow[d, "\mathrm{Id}"] \\ & \langle 1 \rangle \end{tikzcd}$$

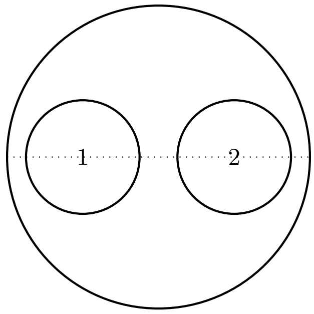
with c_1 and c_2 being configurations in $\mathrm{E}_2(n)$, commutativity means choosing a path from c_1 to c_2. Let β denote the following morphism in $\mathrm{E}_2^{\otimes}$

$$\begin{aligned} \beta\colon \langle 2 \rangle &\to \langle 2 \rangle \\ 1 &\mapsto 2 \\ 2 &\mapsto 1. \end{aligned}$$

with the trivial collection $(\mathrm{Id}^2, \mathrm{Id}^1)$. Further, let

$$\mu \colon \langle 2 \rangle \to \langle 1 \rangle$$

be the morphism with the following configuration which we refer to as the basepoint configuration:



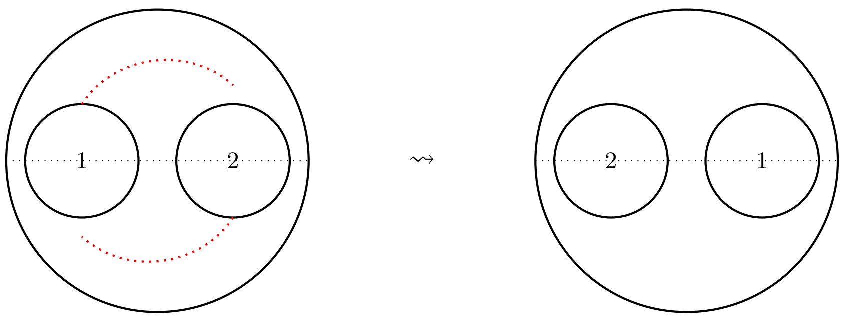
Since π satisfies (M1) and (M2), we have that for any $(A, B) \in \mathcal{C}^2 \cong \mathcal{C}^{\otimes}_{\langle 2 \rangle}$ there is a cocartesian lift $(A, B) \to A \otimes B$ which defines a functor

$$\otimes \colon \mathcal{C}^2 \to \mathcal{C}.$$

Then, we consider the following diagramm in $\mathrm{E}_2^{\otimes}$

$$\begin{tikzcd}
\langle 2 \rangle \arrow[r, "\beta"] \arrow[d, "\mu"] & \langle 2 \rangle \arrow[d, "\mu"] \\
\langle 1 \rangle \arrow[r, dashed] & \langle 1 \rangle.
\end{tikzcd}$$

Filling the diagram, i.e. giving commutativity means choosing a path from the basepoint configuration μ to the twisted basepoint configuration $\mu \circ \beta$. We choose this to be the half twist.



Note, that for a different choice of path instead of this half twist, we would obtain in the following a different braiding. So, this choice is essential for our construction. Again using the (M1) property of π we obtain a commuting diagram in $\mathcal{C}^{\otimes}$

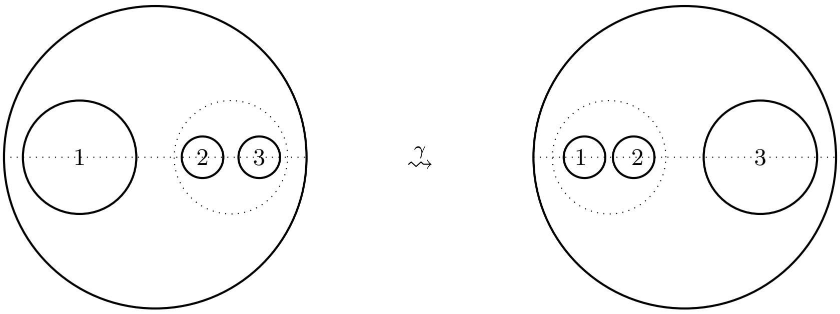
$$\begin{tikzcd}
(A, B) \arrow[r, "\beta"] \arrow[d, "\otimes"] & (B, A) \arrow[d, "\otimes"] \\
A \otimes B \arrow[r, "b"] & B \otimes A
\end{tikzcd}$$

The induced morphism b is a unique isomorphism due to the fact that $\otimes$ and β are cocartesian morphisms.

For the associator we consider the following diagram in $\mathrm{E}_2^{\otimes}$:

$$\begin{tikzcd}
\langle 3\rangle \arrow[r, "\tau_2^3"] \arrow[d, "\tau_1^3"] & \langle 2\rangle \arrow[d] \\
\langle 2\rangle \arrow[r] & \langle 1\rangle
\end{tikzcd}$$

where the associated configurations are either the trivial configuration or the basepoint configuration. To fill the diagram, we choose a path γ that onlz translates the middle points of the disc along the horizontal line and shrinks or expanse the discs as needed.



Again using the (M1) property of π we obtain a commuting diagram in $\mathcal{C}^{\otimes}$

$$\begin{tikzcd}
(A\otimes B, C) \arrow[d, "\otimes"] & (A, B, C) \arrow[l, "(\otimes,\mathrm{Id})"'] \arrow[r, "(\mathrm{Id},\otimes)"] & (A, B\otimes C) \arrow[d, "\otimes"] \\
(A\otimes B)\otimes C \arrow[rr, "a"] & & A\otimes(B\otimes C)
\end{tikzcd}$$

tensor unit is missing

Proposition 2.12. *The category $\mathcal{C}$ equipped with $(\otimes, b, a, \mathbf{1})$ defines a braided monoidal category.*

Proof. First, we observe that the associator refers to E_1 being embedded into E_2 along the horizontal line. So, the pentagon identity holds as E_1 is the associative operad. Hence, we focus on the hexagone identities

include a proof here if spare time

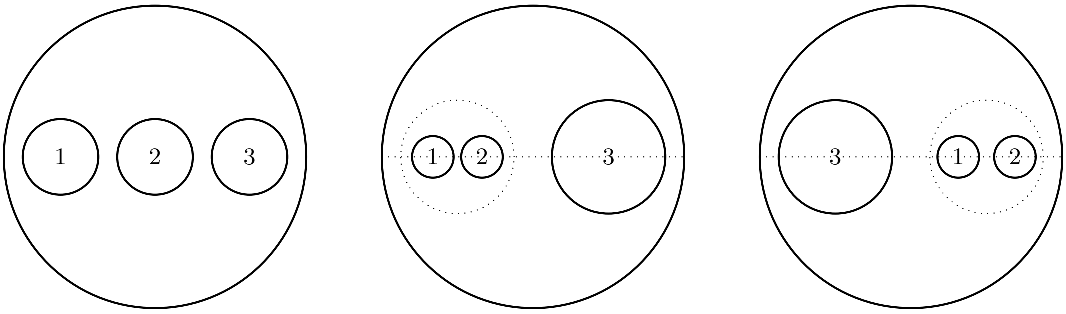
1.
$$\begin{tikzcd}
& A\otimes(B\otimes C) \arrow[r, "b"] & (B\otimes C)\otimes A \arrow[dr, "a"] & \\
(A\otimes B)\otimes C \arrow[ur, "a"] \arrow[dr, "b\otimes\mathrm{Id}"'] & & & B\otimes(C\otimes A) \\
& (B\otimes A)\otimes C \arrow[r, "a"] & B\otimes(A\otimes C) \arrow[ur, "\mathrm{Id}\otimes b"'] &
\end{tikzcd}$$

2.
$$\begin{tikzcd}
& (A\otimes B)\otimes C \arrow[r, "b"] & C\otimes(A\otimes B) \arrow[dr, "a"] & \\
A\otimes(B\otimes C) \arrow[ur, "a"] \arrow[dr, "\mathrm{Id}\otimes b"'] & & & (C\otimes A)\otimes B \\
& A\otimes(C\otimes B) \arrow[r, "a"] & (A\otimes C)\otimes B \arrow[ur, "b\otimes\mathrm{Id}"'] &
\end{tikzcd}$$

Each edge in the hexagone diagrams refers to a commuting diagram in $\mathrm{E}_2^{\otimes}$ of the form

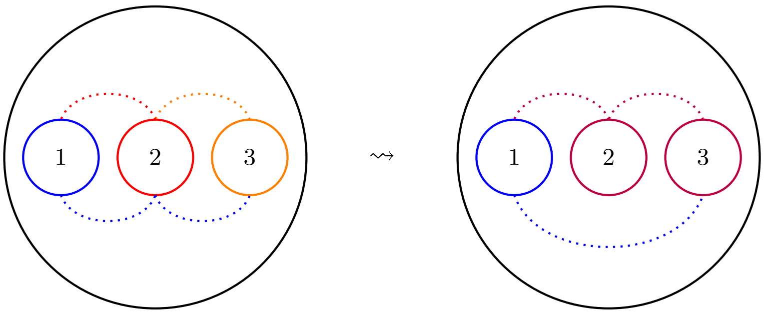
$$
\begin{tikzcd}
& \langle 3\rangle \arrow[ddl, bend right, "c_{A\otimes B\otimes C}"'] \arrow[d, dashed, "c_{(A\otimes B)\otimes C}"] \arrow[ddr, bend left, "c_{C\otimes(A\otimes B)}"] & \\
& \langle 1\rangle \arrow[dl, "\mathrm{Id}"] \arrow[dr, "\mathrm{Id}"'] & \\
\langle 1\rangle \arrow[rr, "\mathrm{Id}"] & & \langle 1\rangle
\end{tikzcd}
$$

with configurations

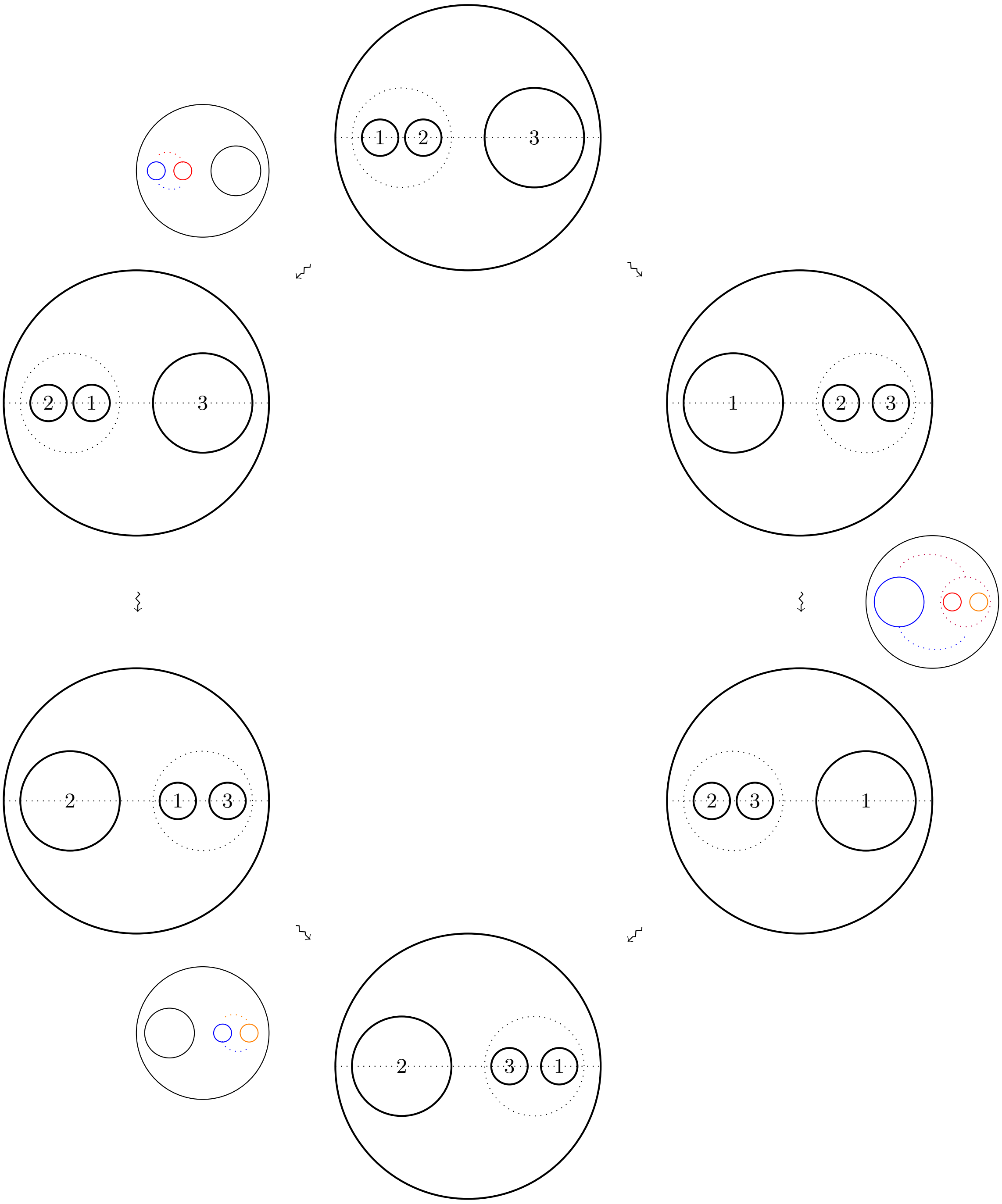


$c_{A\otimes B\otimes C}$ $\quad$ $c_{(A\otimes B)\otimes C}$ $\quad$ $c_{C\otimes(A\otimes B)}$

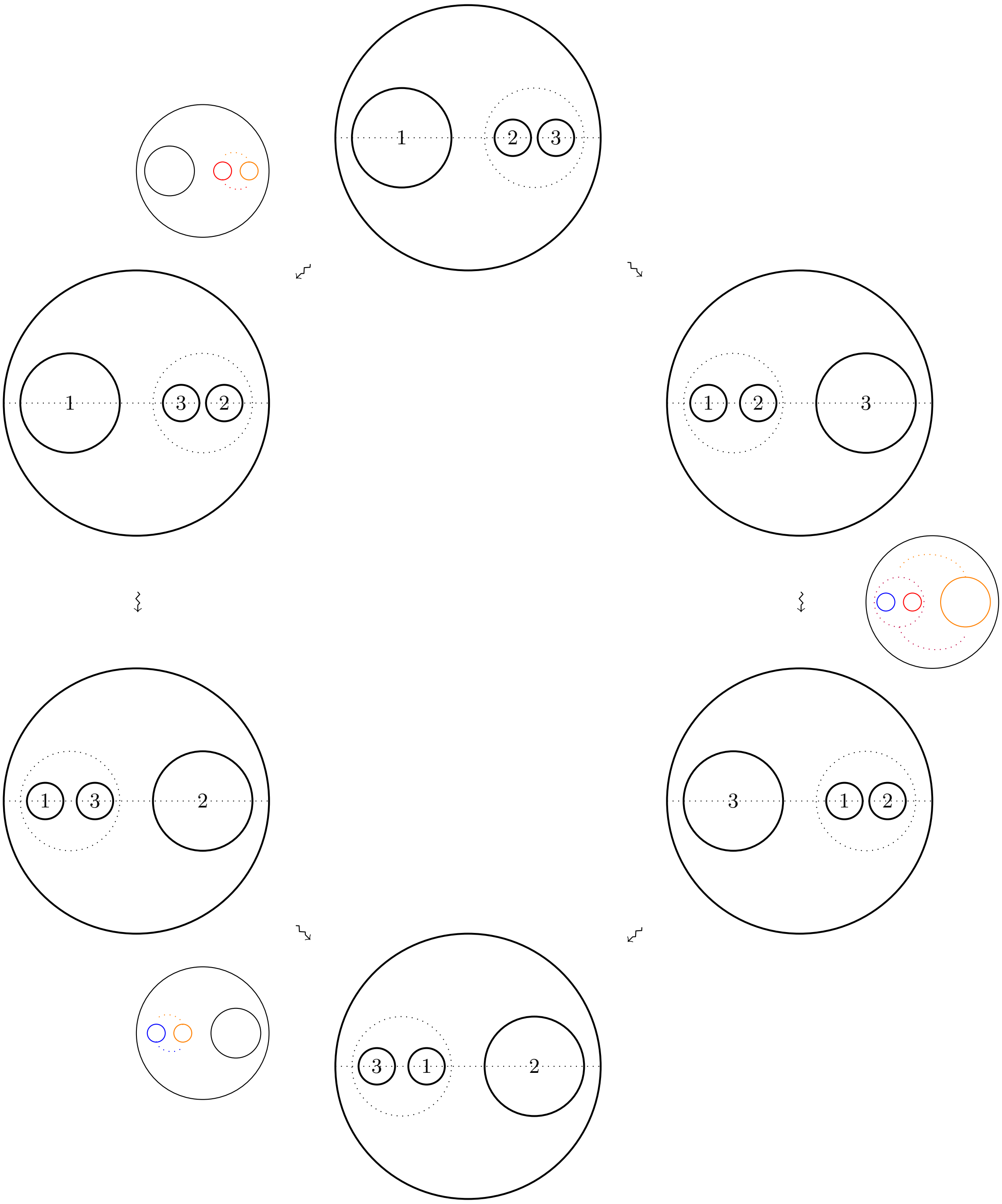
The commutativity of the triangle diagrams "on the side" mean choosing a path from one configuration to another while filling the tetrahedron in $E_2^{\otimes}$ means having a homotopy between the the different paths. As it turns out we do not make any new choices of paths but the paths are induced by the half twist we chose for $\otimes$. The paths are chosen such that the configurations are homotopic to the basepoint configuration.



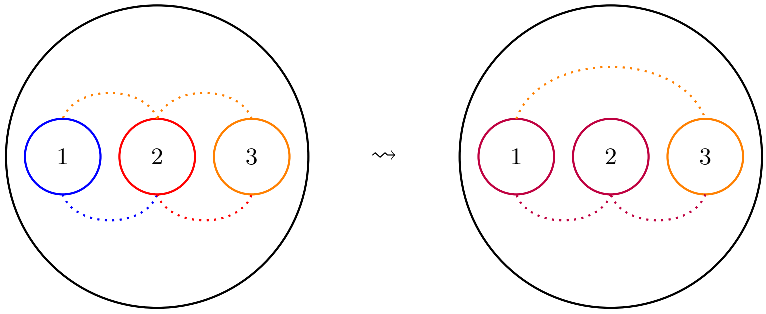
The paths described above follow from the following diagram which is the translation of the first hexagon identity to the corresponding configuration.



For the second hexagon identity we consider the following diagram in $E_2^{\otimes}$:



The resulting paths are again homotopic equivalent and therefore the second hexagon identity holds.



This concludes the proof that $\mathrm{E}_2^{\otimes}$ induces a braided monoidal structure on $\mathcal{C}$.

$\square$

Bibliography

[DK15] T. Dyckerhoff and M. Kapranov. *Crossed simplicial groups and structured surfaces.* 2015.

[FZ] Z. Fiedorowicz. *Constructions of E_n Operads.* 1999.

[KM95] Igor Křížand J. P. May. *Operads, algebras, modules and motives.* en. Astérisque 233. Société mathématique de France, 1995.

[LJ] Jacob Lurie. "Higher algebra". Unpublished. Available online at `https://www.math.ias.edu/~lurie/`. Sept. 2017.

[SC] Christoph Schweigert. "Hopf algebras, quantum groups and topological field theory". Unpublished. Available online at `https://www.math.uni-hamburg.de/home/schweigert/skripten/hskript.pdf`. Mar. 2023.

[YD] Donald Yau. *Infinity Operads and Monoidal Categories with Group Equivariance.* WORLD SCIENTIFIC, 2021.